GEK-SA-1055A

Boiler Stress Monitoring
for the Sabiya Power Station
User Guide



GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. The information is supplied for informational purposes only, and GE makes no warranty as to the accuracy of the information included herein. Changes, modifications, and/or improvements to equipment and specifications are made periodically and these changes may or may not be reflected herein. It is understood that GE may make changes, modifications, or improvements to the equipment referenced herein or to the document itself at any time. This document is intended for trained personnel familiar with the GE products referenced herein.

GE may have patents or pending patent applications covering subject matter in this document. The furnishing of this document does not provide any license whatsoever to any of these patents.

GE provides the following document and the information included therein as is and without warranty of any kind, expressed or implied, including but not limited to any implied statutory warranty of merchantability or fitness for particular purpose.

For further assistance or technical information, contact the nearest GE Sales or Service Office, or an authorized GE Sales Representative.

Revised: May 2013
Issued: Nov 2012

© 2012 - 2013 General Electric Company, All rights reserved.

_____

* Indicates a trademark of General Electric Company and/or its subsidiaries.
All other trademarks are the property of their respective owners.

We would appreciate your feedback about our documentation.
Please send comments or suggestions to controls.doc@ge.com

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Safety Symbol Legend



Warning

Indicates a procedure, condition, or statement that, if not strictly observed, could result in personal injury or death.



Caution

Indicates a procedure, condition, or statement that, if not strictly observed, could result in damage to or destruction of equipment.



Attention

Indicates a procedure, condition, or statement that should be strictly followed to improve these applications.

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Notes

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Contents

Preface ***7***

1 System Overview ***9***

2 System Communication ***15***

2.1 Communication Between the Controller and Workstation 16

2.2 Communication Between the DXT Calculation Engine and HRSGSTRESS_SVR Workstation 17

2.3 Data Transfer Between the Controller and DXT Calculation Engine 20

2.4 Historian Communication 23

3 DXT Calculation Engine ***25***

3.1 Access 26

3.1.1 Full Access 26

3.1.2 Partial Access 26

3.1.3 Monitor Access 26

3.2 Start Up 27

3.3 Software Screens 28

3.3.1 Project Data 28

3.3.2 PT-Monitor 28

3.3.3 Stress Monitor 29

3.3.4 View Menu 30

3.4 Tag Name Mapping Screens 30

3.4.1 Live2csv 30

3.4.2 DXT Mapping 32

3.4.3 Csv2live 33

3.4.4 DXT–Variables 34

3.4.5 Overview Tab 35

3.4.6 All Tag Names Tab 37

3.4.7 Status Tab 38

3.5 Program Settings 39

3.5.1 International Tab 39

3.5.2 Default Values Tab 41

3.5.3 Directories Tab 42

4 HMI Screens and Operator Interface ***45***

4.1 Communication Status 46

4.2 HRSG Operational Cycle Data 47

4.3 Limiting Component 48

4.4 Faceplates 49

5 Historical Data ***51***

6 Diagnostics and Alarms ***53***

6.1 Process Alarms 55

6.2 Component Damage Factors 58

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Notes

GE Proprietary and Internal (Class II) – This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.

Preface

This document provides the low level operation and basic configuration of the GE Boiler Stress Monitoring product as applied at the Sabiya Power Station. Interaction with the product and use of the product outputs are also discussed.

The Boiler Stress Monitoring (BSM) product allows maintenance and operating personnel to manage Heat Recovery Steam Generator (HRSG) operation. It should not be used as a replacement to the recommendations and limits set forth in the operational and maintenance instructions for the equipment provided. The BSM product should not replace the proper inspection and maintenance of HRSG equipment by properly trained, qualified operators and maintenance personnel.

This manual does not provide any guidance on interpretation of stress and life consumption data. All accumulated life data should be considered relative and not absolute. Consult a qualified expert for the proper interpretation and use of the data. GE assumes no liability with respect to accuracy of the data reported against useful life of the equipment provided beyond the obligations set forth in the contracts.

Terms

Parameter or Keyword Units	Description
Creep Damage	Long term exposure to high temperature
Creep Stress	Time-dependent deformation at elevated temperature and constant stress
Danish Xergy Technology (DXT)	Supplier of the EN 12952 compliant calculation engine
EN12952-4	European Standard for Water-Tube Boilers and Auxiliary Installations Part 4: In-Service Boiler life expectancy calculations
Ethernet Global Data (EGD)	Producer/consumer protocol that provides data exchange between nodes at a nominal frame rate of 25 Hz
Fatigue Damage	Structural damage due to exceeding yield strength
ICS	Integrated Control System
Mechanical Stress	Stress due to internal pressures of the various shell and pressure vessels which make up the HRSG
Thermal Stress	Thermal gradients due to thermal transients across thick wall components of the HRSG
Total Damage	Summation of creep and fatigue damage
Unit Data Highway (UDH)	Control layer EGD network for per-to-peer communications

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Notes

GE Proprietary and Internal (Class II) – This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.

1 System Overview

The BSM product is applicable to a wide range of boiler types with varying component geometries and metallurgy and is compatible with all industry standard pressure part design codes.

GE has developed the BSM system as a fully integrated component in the plant Integrated Control System (ICS) to improve the plant operating process and assist in the management of plant assets. The BSM system provides online calculation of thermal, mechanical and creep stress, and corresponding fatigue and creep fractions in steam generator components. This enables plant personnel to evaluate heat recovery steam generator (HRSG) component life consumption against plant operation in a consistent analysis methodology.

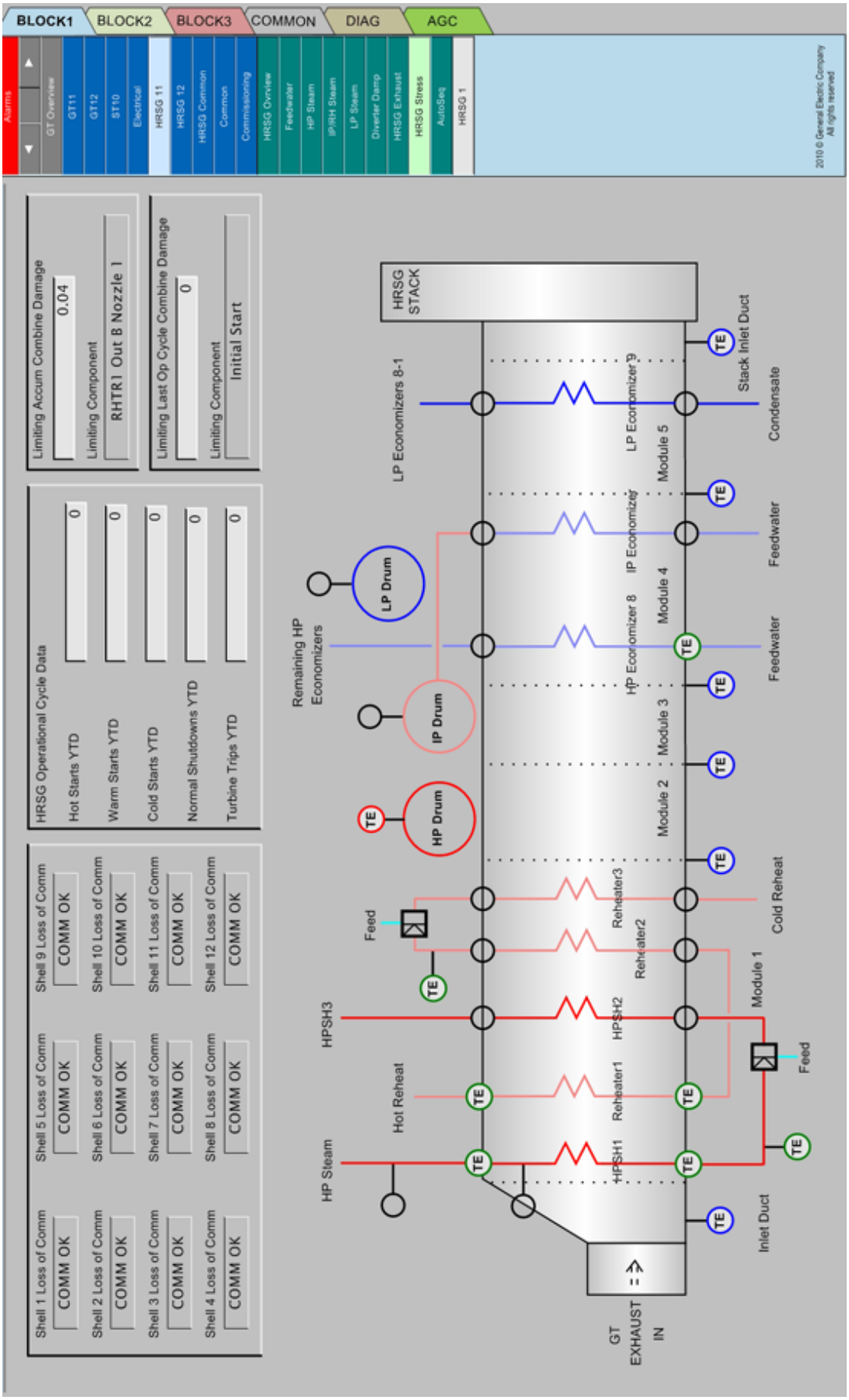
The BSM software applies steady-state methodology of the European Norm standards for the design and dimensioning of shell boilers, with enhancements to allow online, real-time calculations. Coupled with the distributed control system (DCS) and Historian, the monitored component stress levels and fatigue and/or creep damage fractions can be archived for later processing against various plant process data.

Sabiya Power Station

Sabiya Power Station has six HRSGs. Based on GE evaluation, twelve components in the HRSG system were selected as representative of the total HRSG and are monitored by the BSM system. The twelve critical components (drums and shells) include:

- Shell 1 - High Pressure Drum
- Shell 2 - High Pressure Super Heater 1 Outlet Header A
- Shell 3 - High Pressure Super Heater 1 Outlet Header B
- Shell 4 - High Pressure Super Heater 1 Outlet Header C
- Shell 5 - High Pressure Super Heater 1 Inlet Header B
- Shell 6 - Re-Heater 1 Outlet Header A
- Shell 7 - Re-Heater 1 Outlet Header B
- Shell 8 - Re-Heater 1 Outlet Header C
- Shell 9 - Re-Heater 1 Inlet Header B
- Shell 10 - High Pressure Economizer Inlet Header
- Shell 11 - High Pressure Super Heater Attemperator Discharge Pipe
- Shell 12 - Re-Heater Attemperator Discharge Pipe

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



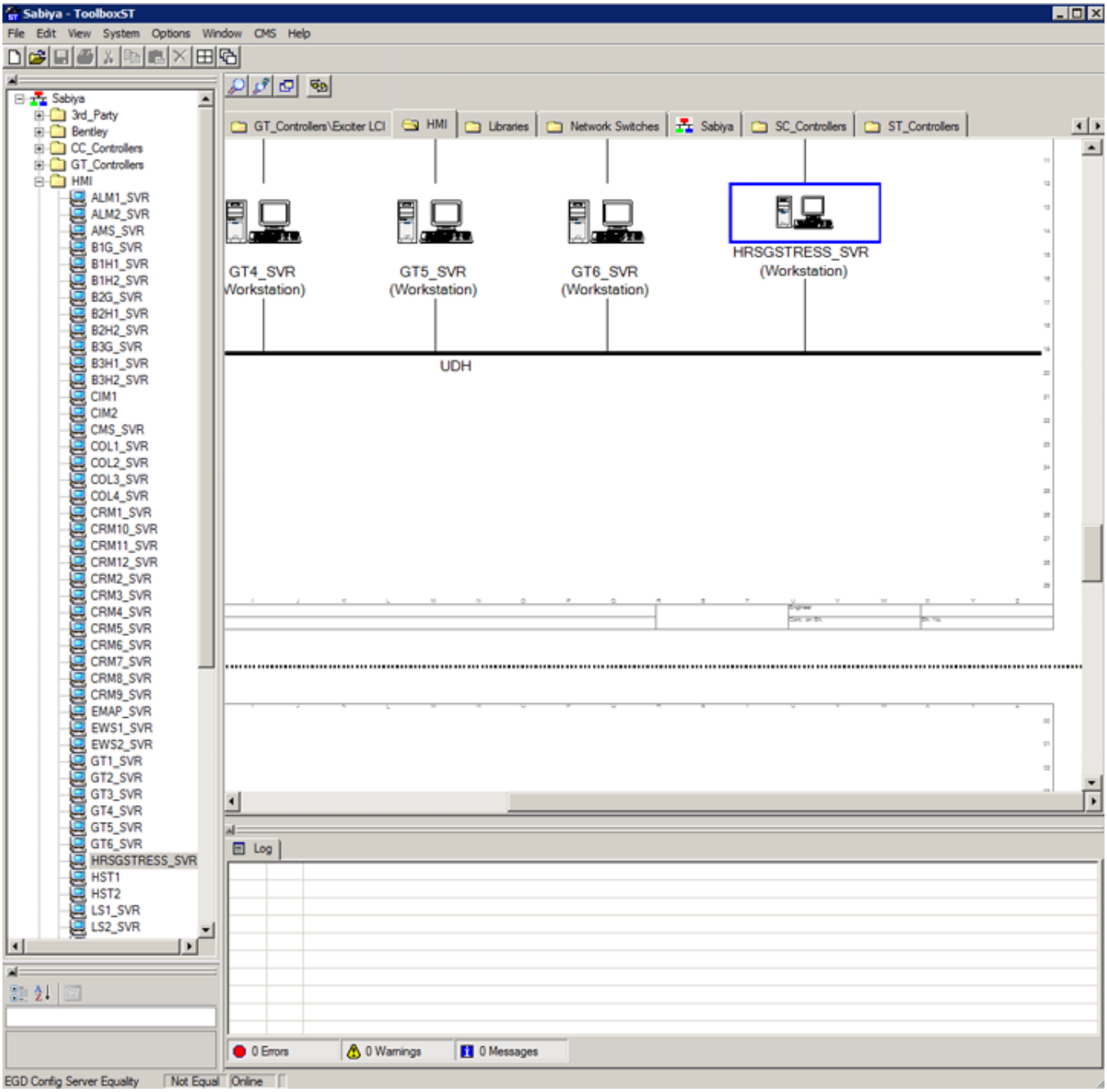
Typical Sabiya BSM Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

The BSM product includes:

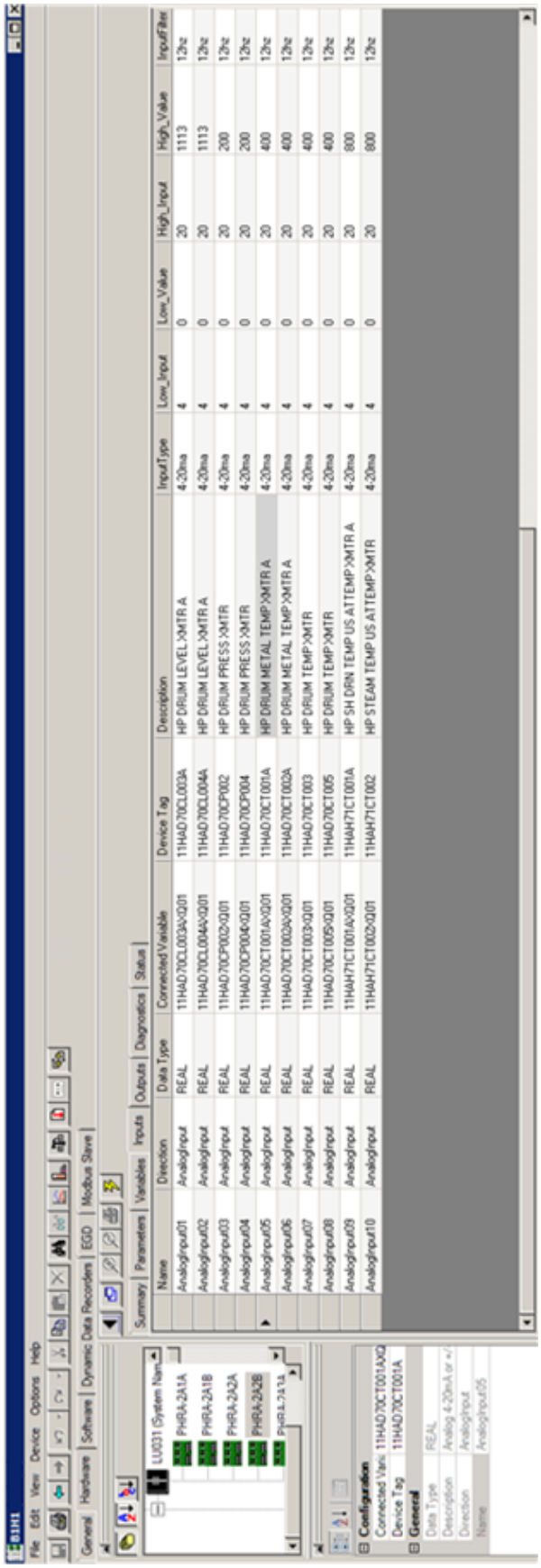
- A dedicated computer workstation, *HRSG_STRESS_SVR*, that is a component of the plant ICS. This workstation has both the WorkStationST* application and the Danish Xergy Technology (DXT) calculation engine. The calculation engine performs the actual stress, fatigue, and creep calculations. The calculation engine interfaces with the Mark* VIe control system through WorkStationST features for the exchange of process and calculated variables. The WorkStationST application allows the workstation to communicate and exchange data with the Mark VIe control system and plant data historian.
- Dedicated thermocouples to measure mid-wall and inner-wall temperature for each component (drum or shell) being monitored. The thermocouples are located in strategic areas of the component, as identified by the HRSG supplier. The temperature gradient in the component, coupled with the temperature rate of change, is critical in calculating the thermal and creep stress.
- A pressure signal for each component is required for both stress and fatigue calculations.
- A Balance of Plant (BOP) controller for each HRSG manages the data exchange with the BSM workstation. This controller serves as the accumulator for the calculated values from the BSM calculations. It also performs diagnostic monitoring and alarming, as well as archiving to the plant data historian.
- The plant data historian is the repository for the accumulated values from each monitored component. This history is a critical data source for analyzing cyclic effects on the HRSG.
- DXT application software adheres to EN 12952 standards and uses rain flow algorithms to assess stress and damage in relation to load cycles. It runs on the dedicated workstation, HRSG_STRESS_SVR.

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



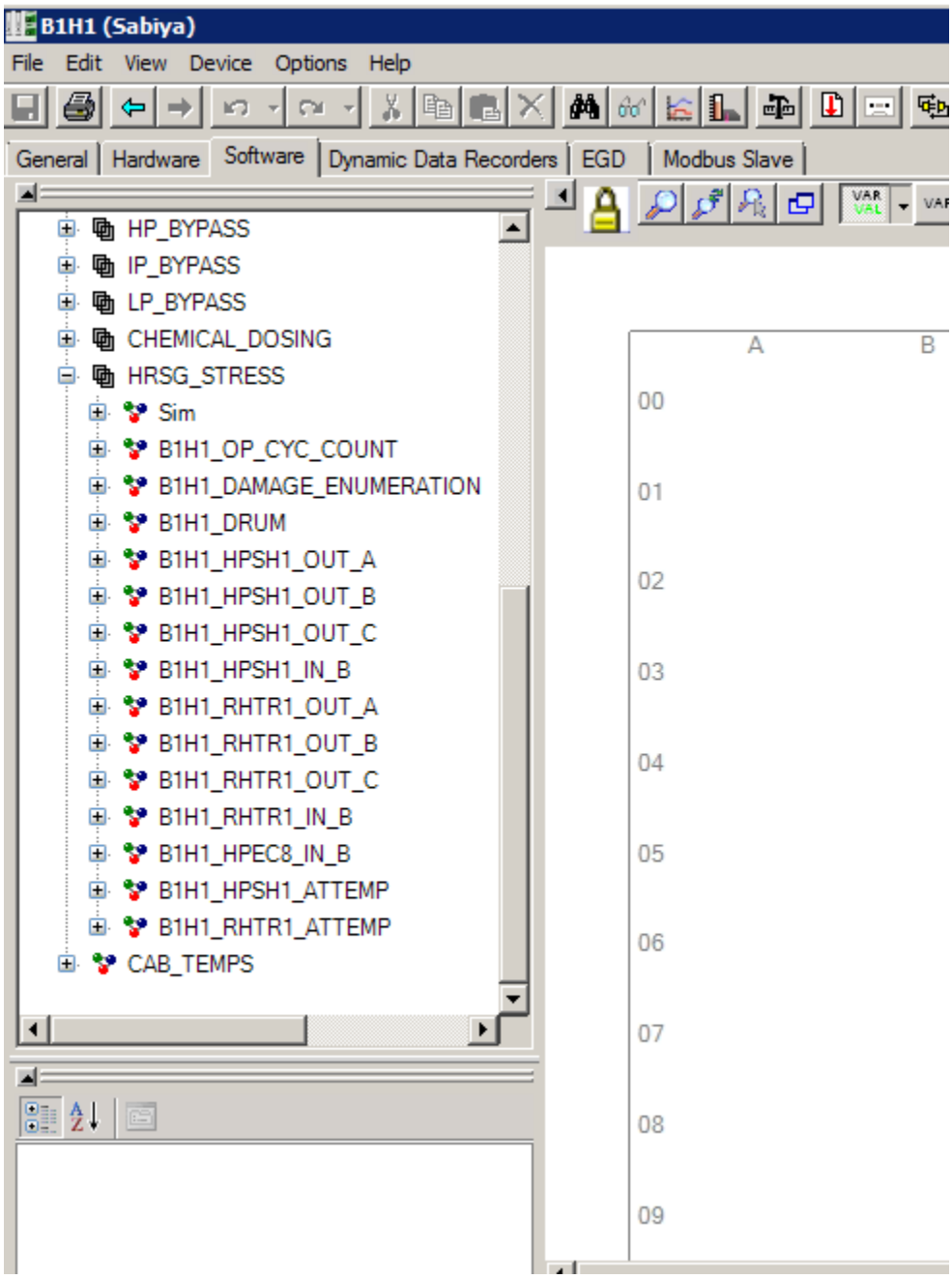
HRSGSTRESS_SVR Component in the Sabiya Mark VIe Control System

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



Example of Component Metal Thermocouples Connecting to the Control System as Standard I/O

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

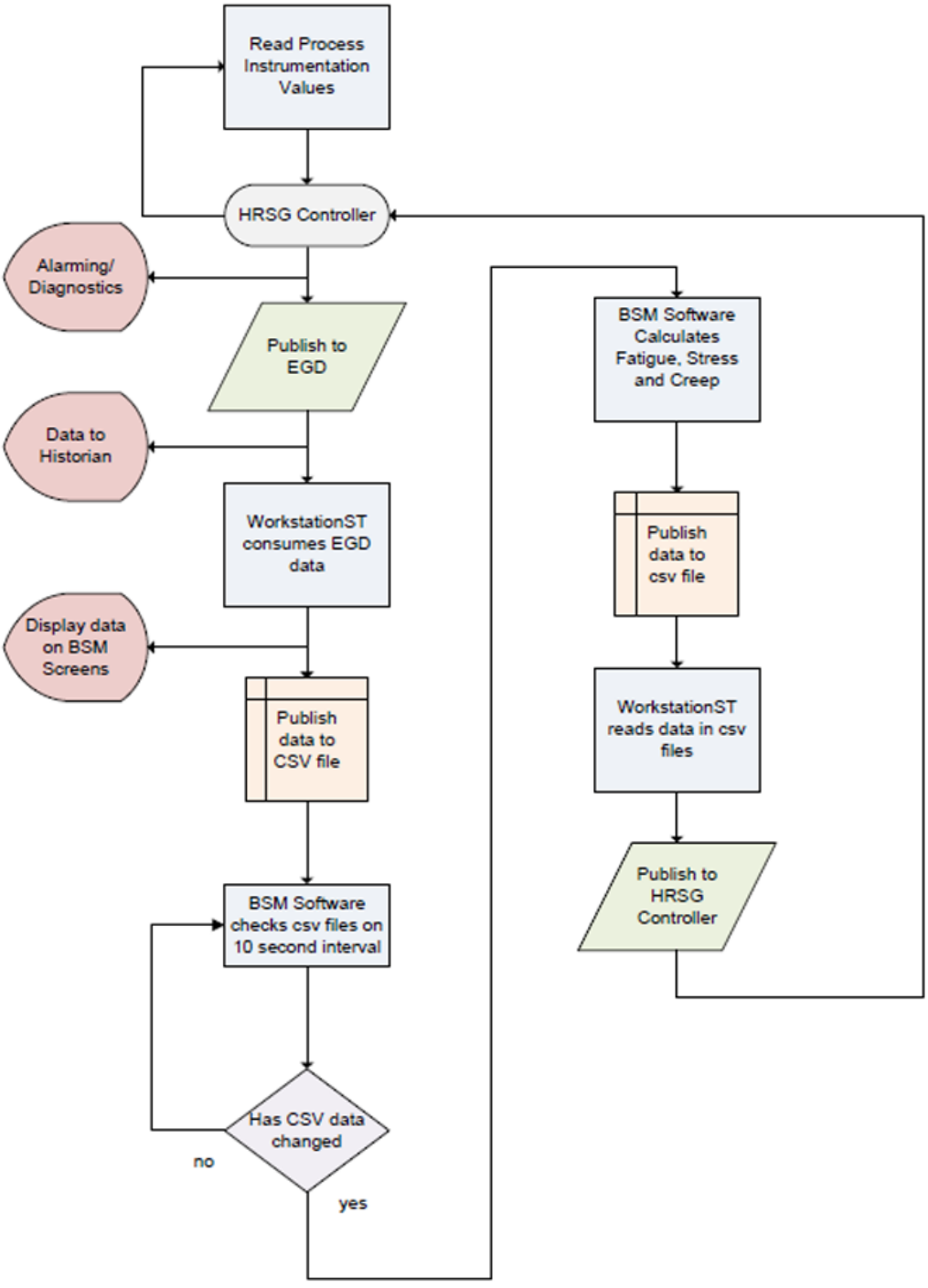


Example of the Location of Controller Logic Block 1 HRSG 1 Controller

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

2 System Communication

The BSM product communication uses various standard Mark VIe communication protocols, supporting communication between the controller and workstation, inner-workstation communication with the DXT online calculation engine, Human-machine Interface (HMI) (operator control interface) screens, and the data plant historian.



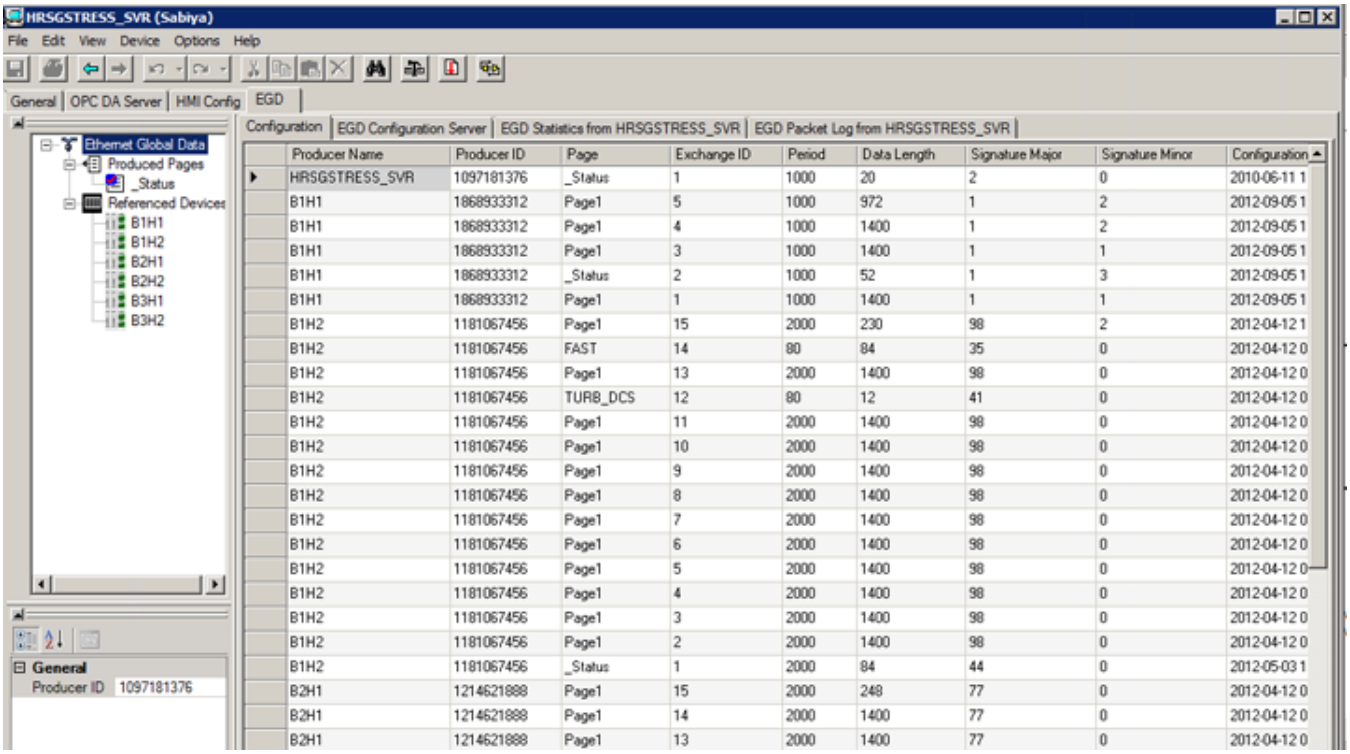
Simplified System Communications

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

2.1 Communication Between the Controller and Workstation

For more information, refer to GEH-6721, Mark VIe Control, Volume 1 System Guide.

The bi-directional data flow between the controller and the workstation leverages GE's Ethernet Global Data (EGD) protocol for efficient data packet exchange. This data transfer uses the Unit Data Highway (UDH) and requires that the HRSG_STRESS_SVR workstation reference each of the block and HRSG controllers. The workstation consumes the EGD pages published by each controller to collect the process data to pass to the DXT calculation engine. The data generated by the DXT calculation engine is passed to the controllers by the workstation using Command Message Protocol (CMP). This is the same protocol used to move operator commands from the HMI screens to the controllers.



Producer Name	Producer ID	Page	Exchange ID	Period	Data Length	Signature Major	Signature Minor	Configuration
HRSGSTRESS_SVR	1097181376	_Status	1	1000	20	2	0	2010-06-11 1
B1H1	1868933312	Page1	5	1000	972	1	2	2012-09-05 1
B1H1	1868933312	Page1	4	1000	1400	1	2	2012-09-05 1
B1H1	1868933312	Page1	3	1000	1400	1	1	2012-09-05 1
B1H1	1868933312	_Status	2	1000	52	1	3	2012-09-05 1
B1H1	1868933312	Page1	1	1000	1400	1	1	2012-09-05 1
B1H2	1181067456	Page1	15	2000	230	98	2	2012-04-12 1
B1H2	1181067456	FAST	14	80	84	35	0	2012-04-12 0
B1H2	1181067456	Page1	13	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	TURB_DCS	12	80	12	41	0	2012-04-12 0
B1H2	1181067456	Page1	11	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	10	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	9	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	8	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	7	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	6	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	5	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	4	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	3	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	Page1	2	2000	1400	98	0	2012-04-12 0
B1H2	1181067456	_Status	1	2000	84	44	0	2012-05-03 1
B2H1	1214621888	Page1	15	2000	248	77	0	2012-04-12 0
B2H1	1214621888	Page1	14	2000	1400	77	0	2012-04-12 0
B2H1	1214621888	Page1	13	2000	1400	77	0	2012-04-12 0

Referenced Controllers for HRSGSTRESS_SVR

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

2.2 Communication Between the DXT Calculation Engine and HRSGSTRESS_SVR Workstation

For more information, refer to GEI-100621, WorkstationST OPC DA Server Installation Guide.

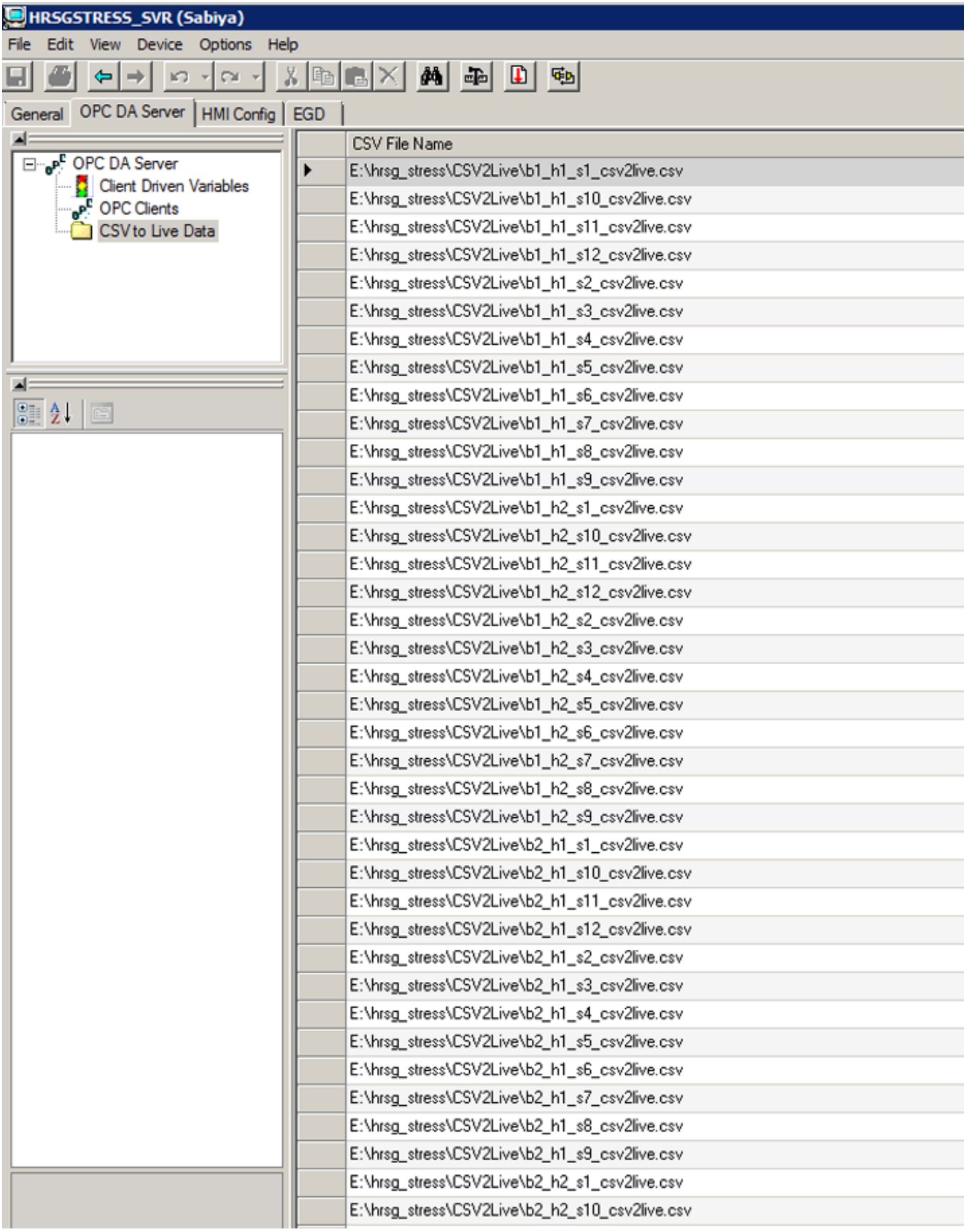
The CSVto Live Data sub-feature of the OPC DA Server brings data generated by the DXT calculation engine to the workstation. The CSVto Live Data feature is designed to move data contained in properly configured Comma Separated (.csv) files into the workstation OPC DA server and to have it sent to the controllers through Command Message Protocol.



Caution

The .csv files should never be modified. The format and time stamp of the file is critical to the program's ability to uptake the data.

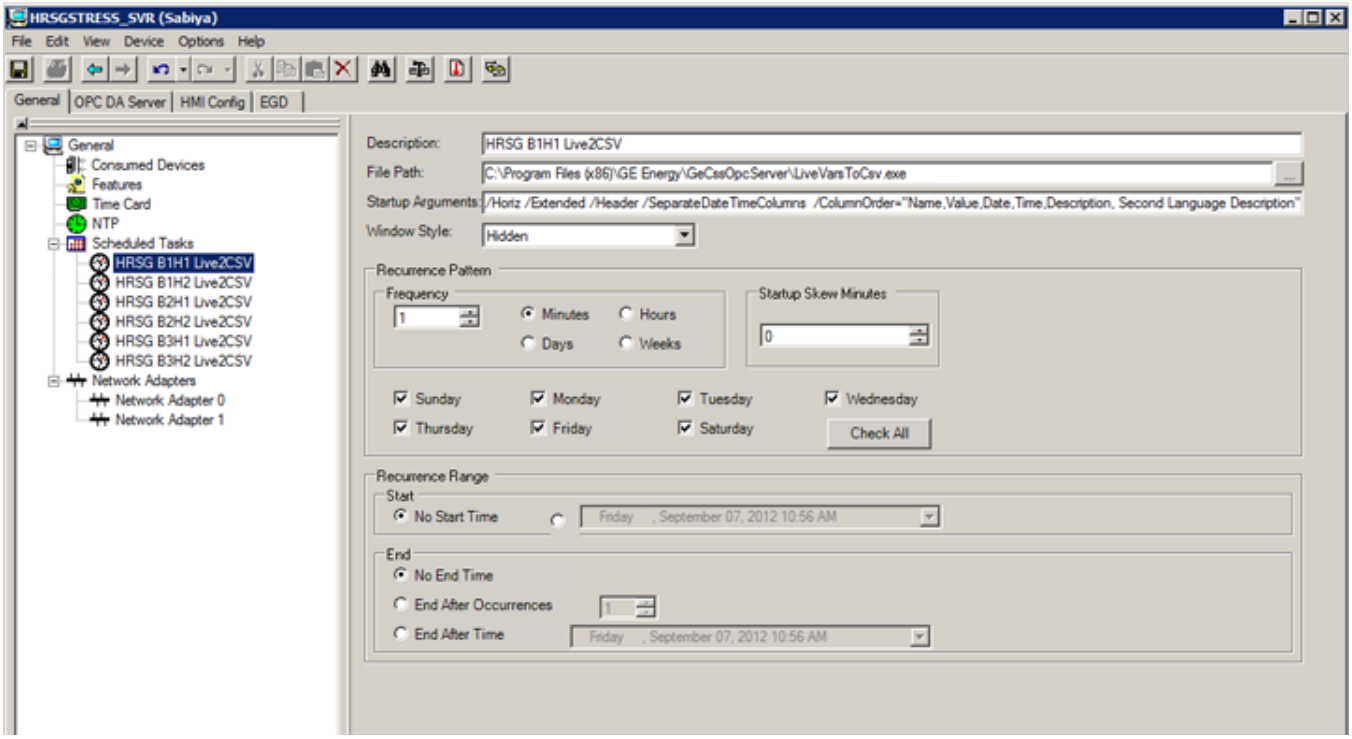
GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



Configured File Structure for the CSV to Live Data Transfer in the HRSGSTRESS_SVR

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

The LiveVarsToCSV sub-feature transfers process data to the online software using WorkstationST controlled schedule tasks. For Sabiya, the HRSGSTRESS_SVR has six scheduled tasks, one for each HRSG. These scheduled tasks operate at a frequency of one per minute and this frequency sets the time-base for the online calculations.



Scheduled Tasks in the HRSGSTRESS_SVR Workstation

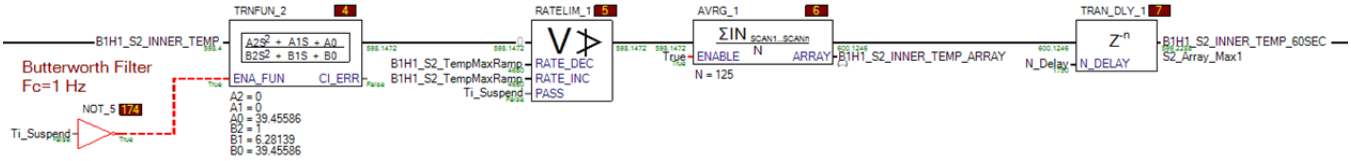
GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

2.3 Data Transfer Between the Controller and DXT Calculation Engine

The metal temperature and vessel pressure signals originate as 4–20 mA analog signals from a field-mounted transmitter and are connected to Mark* VIe ICS inputs. The controller receiving the signals performs filtering and data quality testing before passing the signals through the workstation to the DXT online calculation engine.

The filtering component comprises a Butterworth filter with a 1 Hz cutoff frequency (fc) to filter out high frequency noise. The Butterworth filter is followed by a rate limiter that has the rate cutoff above the maximum rate that the actual process can change. The rate limiter helps to filter the zone above real data and under 1 Hz. The rate limiter is followed by a 10 second rolling average operating at frame rate. At frame rate, the 10 second frame rate is a rolling average of 125 samples (10 seconds/0.08 second frame rate). The rolling average helps filter out low level sinusoidal noise.

The final filtering is the passing of the signals through a data buffer to delay the signal by 60 seconds before being published to the workstation. The data buffer also operates at frame rate so no additional filtering occurs. The 60 second delay allows more than adequate time for the DCS to detect data quality issues and send the appropriate calculation command to the DXT online calculation engine to ensure no calculations are performed with bad quality data.



Standard Process Signal Filtering Scheme

Data quality testing includes checking for bad quality status as detected by the input circuitry of the DCS, metal temperatures greater than the maximum scaled 4–20 mA of the transmitter or the calculated temperature maximum of the metal in the component as calculated by DXT for the online calculation engine, temperature below 5°C (41 °F), unless the HRSG is offline, then the temperature setpoint moves to 0°C (32 °F). When a bad quality status is detected on either of the metal temperatures or pressure for a component, the suspend variable originating in the DCS is appropriately updated to signal the online calculation engine to suspend calculations that use the signals tagged as bad quality. A suspend variable exists for each component monitored on each HRSG. The suspend variables have the format *BxHx_Sx_Suspend*.

The data output from the DXT online calculation engine includes the following important variables, which are calculated and updated every minute:

- FILE_EX — Verifies that communications are active and provides feedback that the online software has received and responded to a GT halt or signal bad quality calculation suspend command.
- BxHx_Sx_NZx_CSCMS — Calculated Creep Membrane Stress
- BxHx_Sx_NZx_CSDF — Creep Damage Factor
- BxHx_Sx_NZx_FSCDF — Fatigue Damage Factor
- BxHx_Sx_NZx_FSCMS — Calculated Mechanical Fatigue Stress

GE Proprietary and Internal (Class II) – This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.

- BxHx_Sx_NZx_FSCTS — Calculated Thermal Fatigue Stress

Original Fill Exchange Counts - Superceded by Matrix Below				
			File Exchange Count	
Tref	Creep	Fatigue	Min	Max
x	x	x	1	100000
0	0	0	-1	-9
x	0	x	-10	-19
x	x	0	-20	-29
0	x	x	-30	-39
0	0	x	-40	-49
0	x	0	-50	-59
x	0	0	-60	-69

Tref - Refernce Temperature	x	Parameter valid
	0	Parameter not valid

FILE_EX Variable Feedback Matrix to DCS.

				Bad signal on						
GT-status	Suspend	File exchange count.	GT running	Ti	Tm	TiAW	TmAW	TiBW	TmBW	P
1	0	1 to 100000	Yes	NA	NA	NA	NA	NA	NA	NA
0	0 to 7	-70 to -79	No	NA	NA	NA	NA	NA	NA	NA
1	1	-80 to -89	Yes	0	x	x	x	x	x	x
1	2	90 to -99	Yes	x	0	x	x	x	x	x
1	3	-100 to -109	Yes	x	x	0	x	x	x	x
1	4	-110 to -119	Yes	x	x	x	0	x	x	x
1	5	120 to -129	Yes	x	x	x	x	0	x	x
1	6	-130 to -139	Yes	x	x	x	x	x	0	x
1	7	140 to -149	Yes	x	x	x	x	x	x	0

Count recycles the range as long as the condition exists

Ti - Temperature Inner	x	Parameter valid
Tm - Temperature Middle	0	Parameter not valid

TiAW - Temperature Inner Above Water

TmAW -Temperature Middle Above Water

TiBW - Temperature Inner Below Water

TmBW -Temperature Middle Below Water

P - Pressure

FILE_EX Variable Possible Values

A two hour delay allows for uniform cooling of the components of the HRSG.

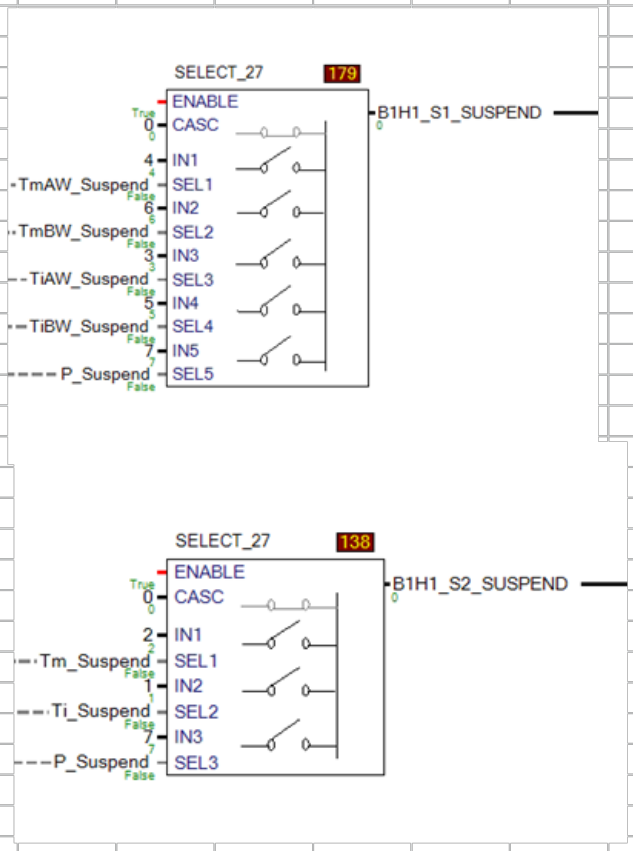
The GT status variable takes the format, *BxHx_GT_STATUS*, and indicates that the gas turbine for the particular HRSG is either firing or rotating at purge speed or greater and the diverter damper and Guillotine are open to allow exhaust gas flow through the HRSG. When exhaust flow is present in the HRSG, the GT status is set to an integer value of one. On shut down where the gas turbine speed is under 10%, or any condition where the exhaust flow is not through the HRSG, a two hour delay will follow before the GT status is set to 0. After two hours, the temperature difference between inner and middle temperatures will have reached equilibrium. When the online calculation receives a GT status of 0, the calculations will halt for all components of the affected HRSG. The calculation engine will close all open cycles associated with its calculations to ensure that no matter how long the HRSG is down, the online calculation engine will start new calculations on a HRSG restart.

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

HP Drum	
BxHx_S1_SUSPEND	
Bad Quality	*Integer Value
TmAW_Suspend	**4**
TmBw_Suspend	**6**
TiAW_Suspend	**3**
TiBW_Suspend	**5**
P_Suspend	**7**
Shells	
BxHx_Sx_SUSPEND	
Bad Quality	*Integer Value
Tm_Suspend	**2**
Ti_Suspend	**1**
P_Suspend	**7**

*** Listed in order of priority**

TiAW - Temperature Inner Above Water
TmAW -Temperature Middle Above Water
TiBW - Temperature Inner Below Water
TmBW -Temperature Middle Below Water
Ti - Temperature Inner
Tm -Temperature Middle
P - Pressure

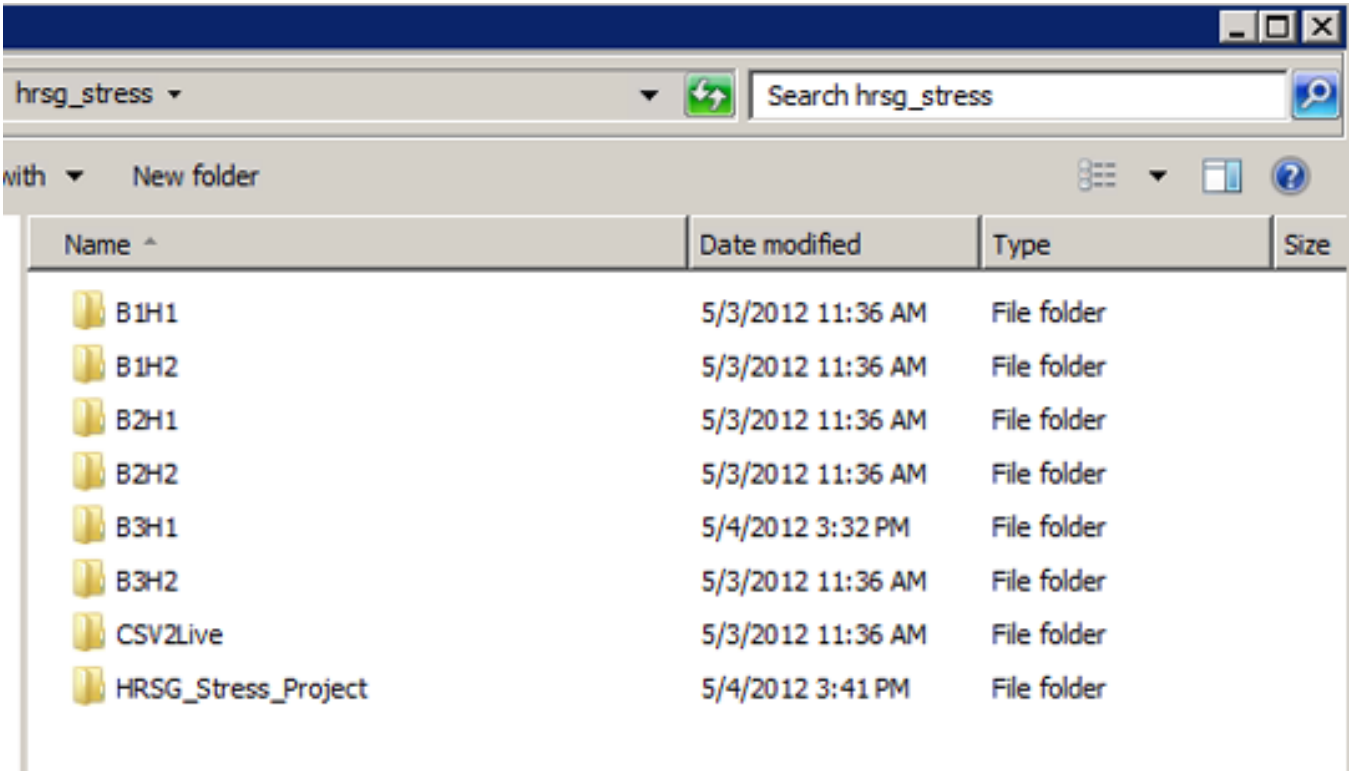


BxHx_Sx_SUSPEND Variable Possible Values

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

2.4 Historian Communication

All data sent to the plant data historian originates from each local HRSG controller, B1H1, B1H2, and so forth. Historian tag attributes are adjusted to indicate a historical point and placed on the EGD. The historian collects the data from the referenced device EGD pages, through the OPC DA Server with which the OSI OPC interface communicates, to pass data to the historian.



File Structure of the CSV Files

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Example of HRSG Stress Monitoring Tags Configured for the Historian

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3 DXT Calculation Engine

Creep damage is determined using material creep properties, as well as local stress and exposure time.

The DXT calculation engine calculates parameters for evaluating and determining instantaneous and accumulated fatigue and creep damage factors for the selected components of the HRSG system.

Based on pressure and temperature measurements coming from the DCS, local stresses and stress ranges are calculated and returned to the DCS historian for storage. From the calculated stress and stress range variation, fatigue damage can be determined using the Rain Flow/Range Pair method for cycle counting as described in EN12952-4.

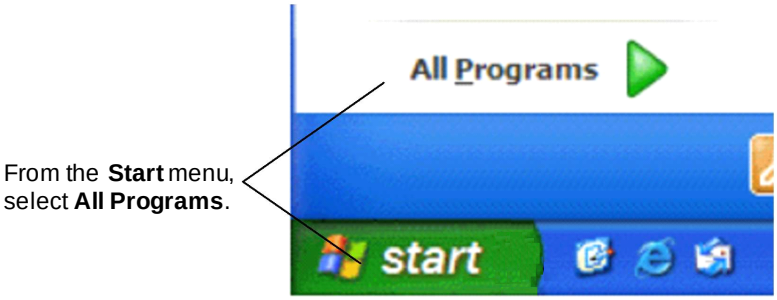
The application is pre-installed on a workstation solely dedicated to the calculation of stress and lifecycle monitoring. Normally, it automatically starts when powering on the workstation. The application can be manually started from the Windows® Start menu.

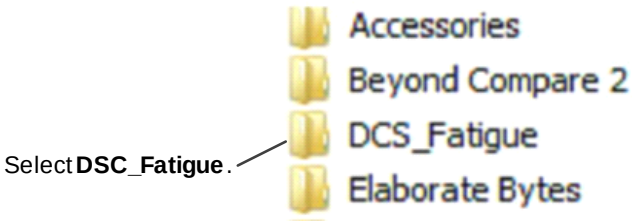


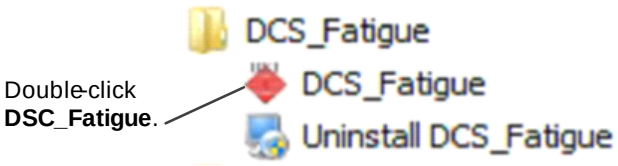
Attention

Before running the program, the HRSG component geometry, material properties, weld type, weld quality, and so forth must be specified. The component configuration is based on detailed drawings and design information from the HRSG vendor and will be made by GE prior to installing the software on site.

➢ **To start the BSM online software**







GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.1 Access

The BSM program runs in three different access modes.

Access Mode	Password Required
Full access	Yes
Partial access	Yes
Monitor access	No

3.1.1 Full Access

Full access mode is restricted to DXT and GE. All program functionality is available to the user for setting up new projects, including:

- Input for component geometry, material, mechanical strength, stress concentration factors, and so forth
- Tag name definition and data exchange setup for communication with the DCS

Expertise is required for configuration. Only a user with full access to the system will be able to view the configuration screens.



Caution

Any modifications can affect the algorithms of the calculation engine, rendering the output of the online software incorrect. Once the proper configuration for the BSM is completed by GE, it must not be altered.

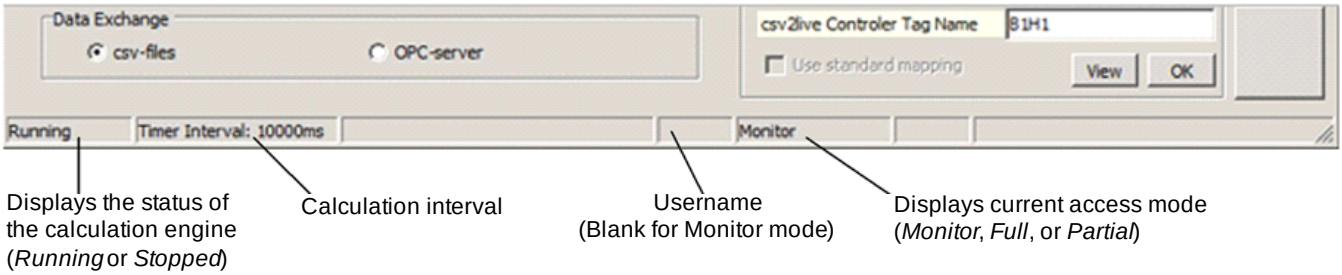
3.1.2 Partial Access

Partial software access is restricted to GE and gives permission for:

- Tag name definition and data exchange setup for communication with the DCS
- Timer value for the frequency of calculations

3.1.3 Monitor Access

Monitor access is the default mode. It allows the user to view individual menus but not change any settings. The status bar in the bottom of the main screen displays the status of the online software.



Monitor Access Main Screen

***GE Proprietary and Internal (Class II)** – This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.2 Start Up

The DXT calculation engine runs in Monitor mode at start up. To change any settings or the HRSG configuration, the user must log on with a username and password provided by GE. The username and password determine the access level permissions available for the user.

Note GE restricts access of the software configuration parameters to GE personnel.

During start up, the configuration data specific to the actual site is automatically loaded and starts the communication with the DCS. Data exchange between the HRSG Stress and LifeCycle Monitor and the DCS can be performed in two ways:

Both communication methods require unique configuration and are not interchangeable on site. Sabiya presently leverages the Live2csv method. Refer to the section, Live2csv.

- Using comma separated files (.csv files).
- Using the WorkstationST OPC DA Server. The online software (DXT) acts as an OPC client.

Key features include:

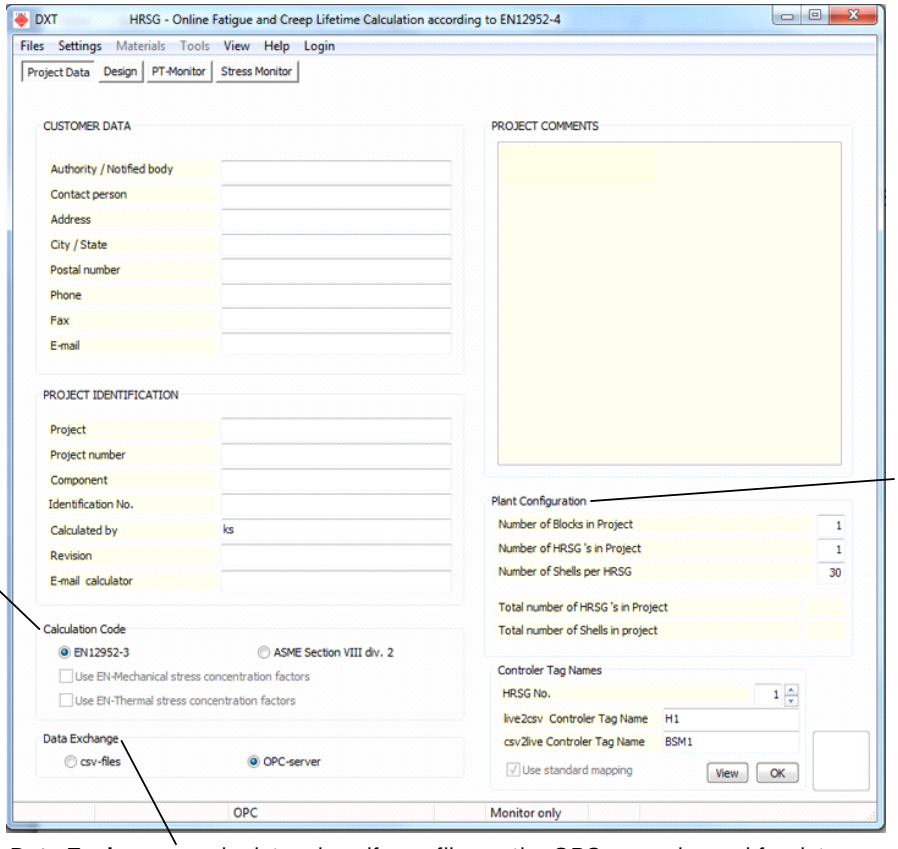
- The actual data exchange is determined by GE before installing the software.
- If the workstation is accidentally shut down, the software starts up and runs in Monitor mode on reboot.
- If the communication between the online software and the DCS is lost, the software will keep trying to connect to the DCS provided .csv files or OPC server.
- The communication status of *Running* or *Stopped* is displayed in the status bar.

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.3 Software Screens

3.3.1 Project Data

The Project Data screen contains information on the actual project setup. Modification of specific parameters are password controlled.



Calculation Code determines if the stress calculation is based on the EN12952-3 standard or the ASME Section VIII div. 2 standard. The EN-12952-2 standard is the default selection.

Data Exchange mode determines if .csv files or the OPC server is used for data exchange between the DCS and BSM software.

Plant Configuration is used for Tag Name mapping to define the data exchange between the DCS and calculation parameters.

Based on the actual numbers of blocks, HRSGs, and shells, the necessary Tag Names will automatically generate using controller tag names. They are listed when the user is logged on with either *Partial* or *Full* access by pressing **View**.

3.3.2 PT-Monitor

The PT-Monitor screen displays a graphical representation of the last 100 measurements of inner and middle wall temperatures and water/steam pressure for each individual monitored component shell. The actual shell can be selected in the menu located below the chart.

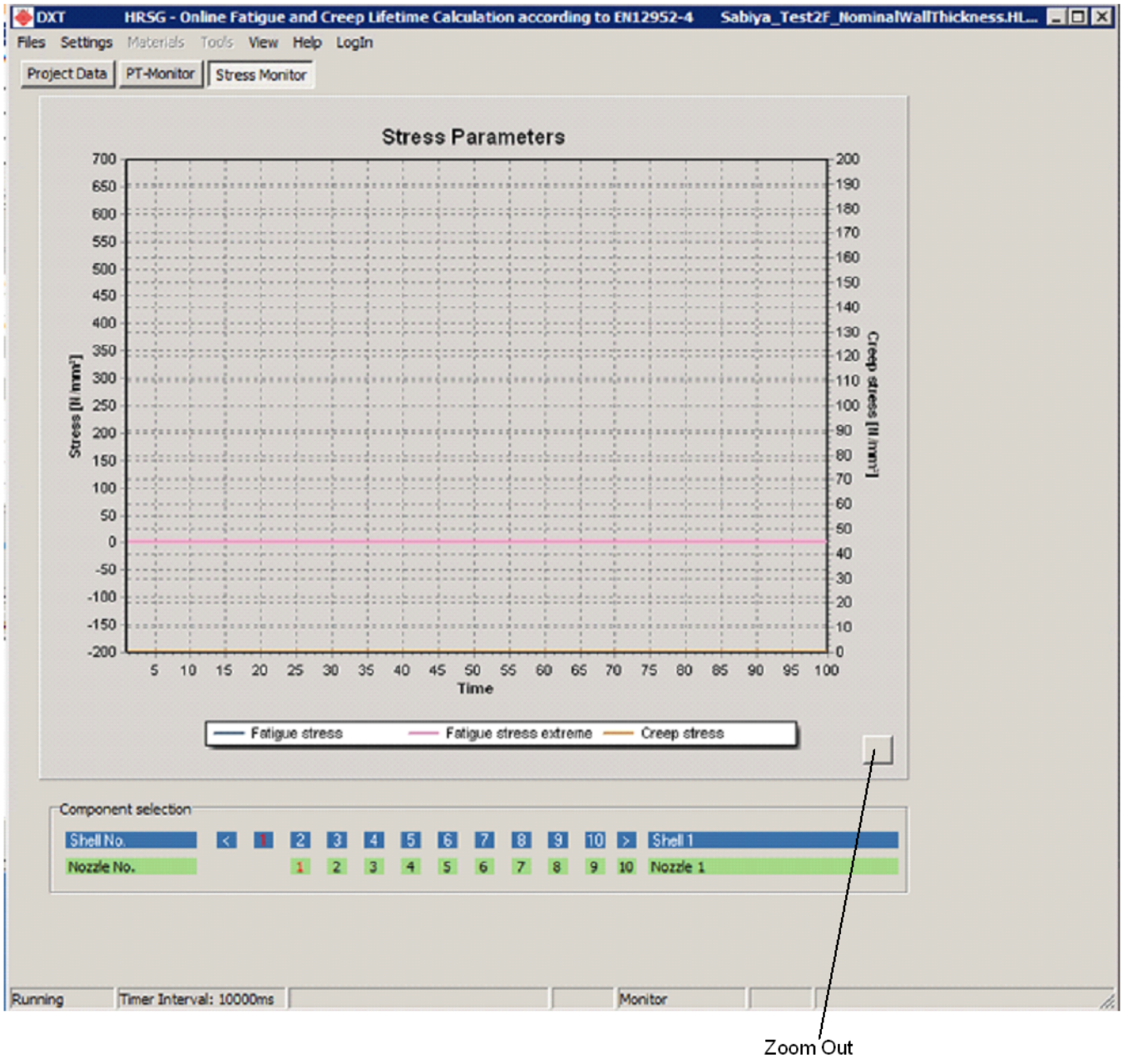
Tip *Zoom in by clicking the left-mouse button while drawing a rectangle on the graph.*

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.3.3 Stress Monitor

The Stress Monitor screen displays the last 100 values for calculated fatigue stress, fatigue stress extreme values, and creep stress for each individual nozzle. The shell or nozzle can be selected in the Component Selection menu.

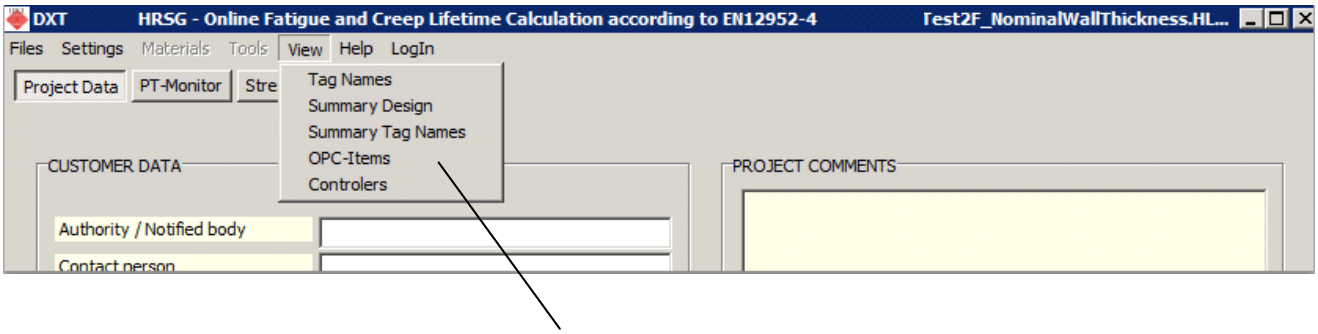
Tip *Zoom in by clicking the left-mouse button while drawing a rectangle on the graph.*



Stress Monitor Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.3.4 View Menu



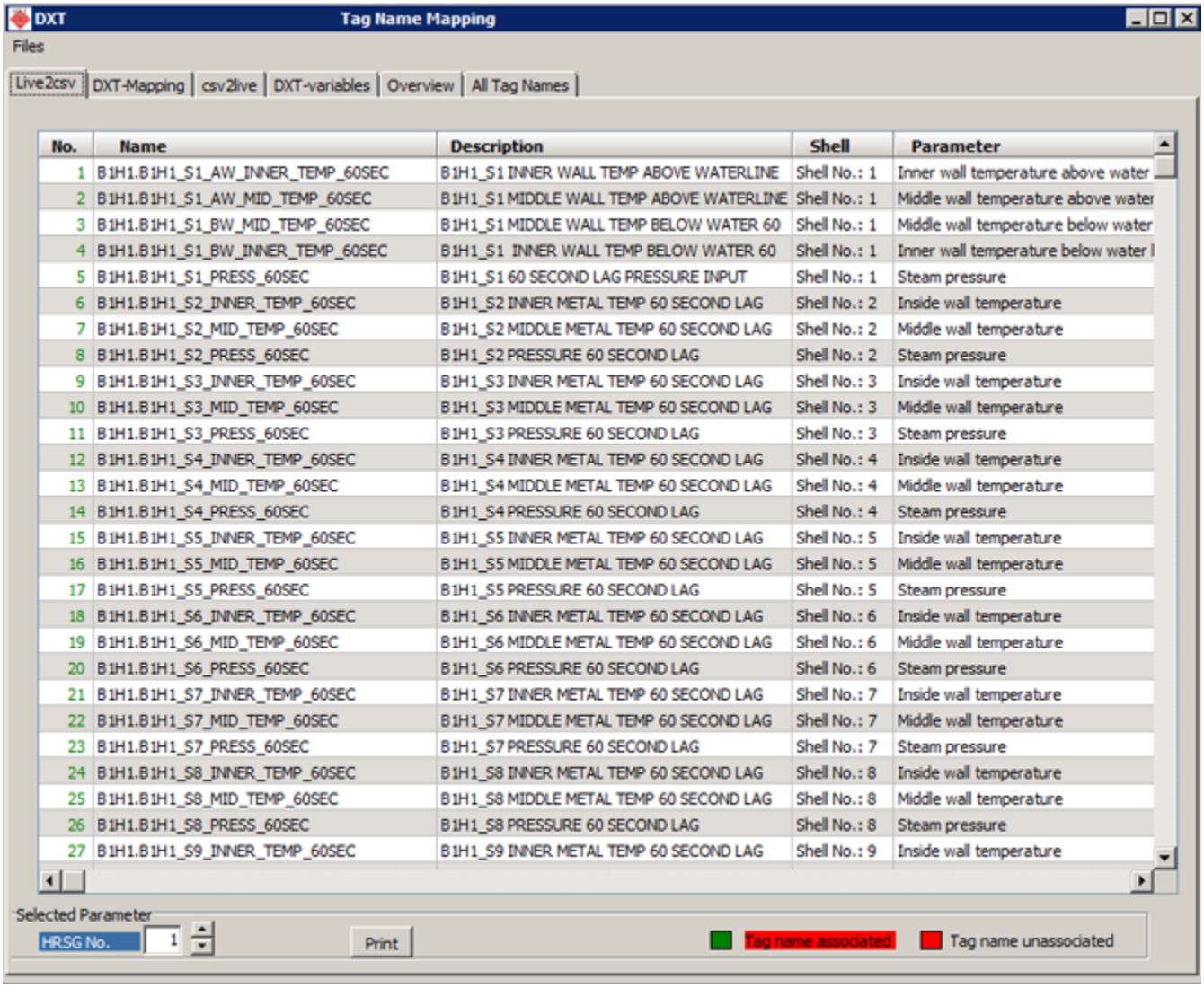
The **View** menu provides access to different design and online parts.

3.4 Tag Name Mapping Screens

3.4.1 Live2csv

The Live2csv screen displays the status for the Tag Name mapping of parameters and values coming from the DCS to the BSM software. A green number in the first column indicates that the actual Tag Name has an association. A red number indicates that no association is found for that specific Tag Name. If the project contains more than one HRSG, different HRSGs can be selected in the Selected Parameter menu. Columns can be scaled by dragging the headings.

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



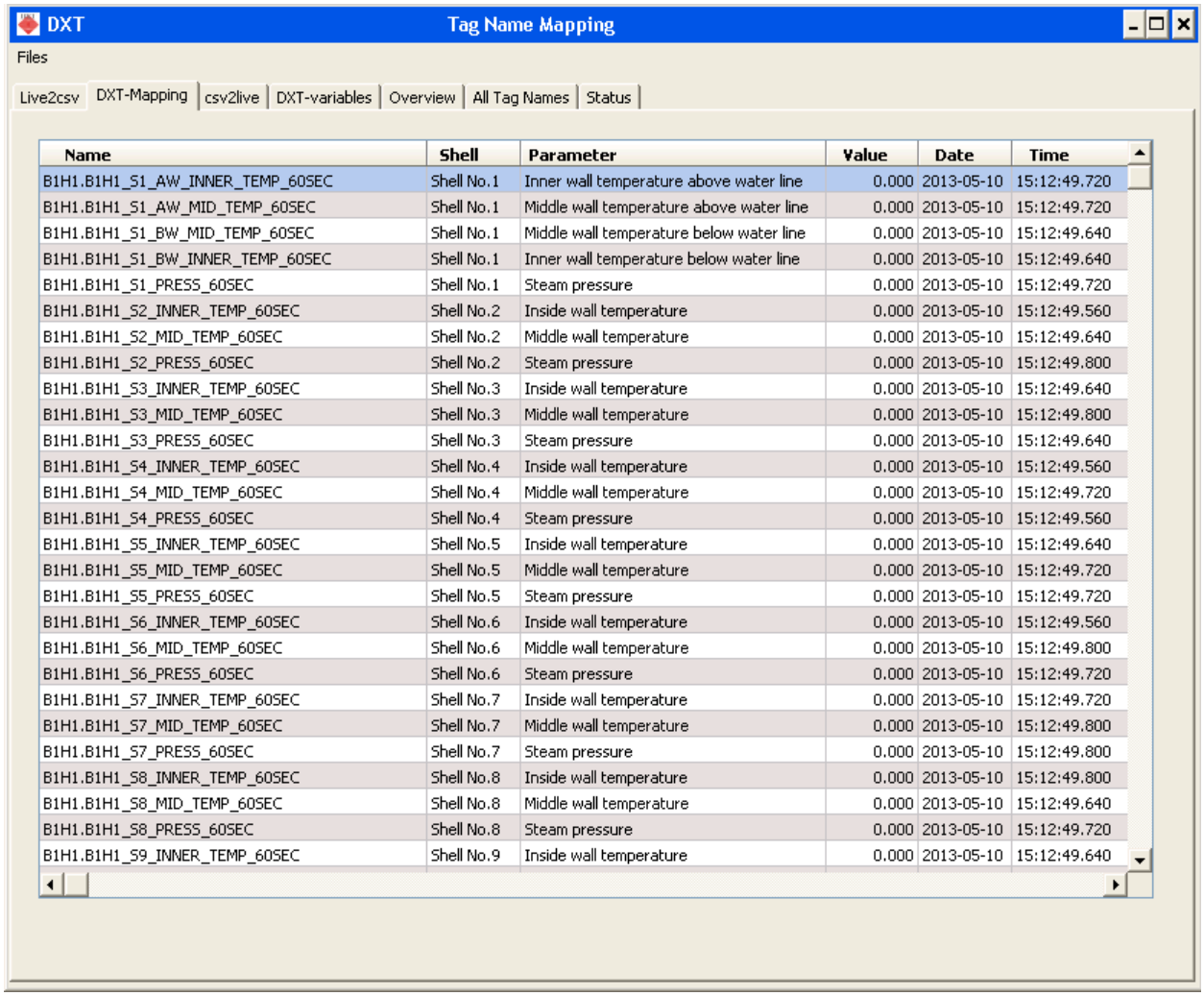
No.	Name	Description	Shell	Parameter
1	B1H1.B1H1_S1_AW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP ABOVE WATERLINE	Shell No.: 1	Inner wall temperature above water
2	B1H1.B1H1_S1_AW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP ABOVE WATERLINE	Shell No.: 1	Middle wall temperature above wate
3	B1H1.B1H1_S1_BW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP BELOW WATER 60	Shell No.: 1	Middle wall temperature below water
4	B1H1.B1H1_S1_BW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP BELOW WATER 60	Shell No.: 1	Inner wall temperature below water l
5	B1H1.B1H1_S1_PRESS_60SEC	B1H1_S1 60 SECOND LAG PRESSURE INPUT	Shell No.: 1	Steam pressure
6	B1H1.B1H1_S2_INNER_TEMP_60SEC	B1H1_S2 INNER METAL TEMP 60 SECOND LAG	Shell No.: 2	Inside wall temperature
7	B1H1.B1H1_S2_MID_TEMP_60SEC	B1H1_S2 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 2	Middle wall temperature
8	B1H1.B1H1_S2_PRESS_60SEC	B1H1_S2 PRESSURE 60 SECOND LAG	Shell No.: 2	Steam pressure
9	B1H1.B1H1_S3_INNER_TEMP_60SEC	B1H1_S3 INNER METAL TEMP 60 SECOND LAG	Shell No.: 3	Inside wall temperature
10	B1H1.B1H1_S3_MID_TEMP_60SEC	B1H1_S3 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 3	Middle wall temperature
11	B1H1.B1H1_S3_PRESS_60SEC	B1H1_S3 PRESSURE 60 SECOND LAG	Shell No.: 3	Steam pressure
12	B1H1.B1H1_S4_INNER_TEMP_60SEC	B1H1_S4 INNER METAL TEMP 60 SECOND LAG	Shell No.: 4	Inside wall temperature
13	B1H1.B1H1_S4_MID_TEMP_60SEC	B1H1_S4 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 4	Middle wall temperature
14	B1H1.B1H1_S4_PRESS_60SEC	B1H1_S4 PRESSURE 60 SECOND LAG	Shell No.: 4	Steam pressure
15	B1H1.B1H1_S5_INNER_TEMP_60SEC	B1H1_S5 INNER METAL TEMP 60 SECOND LAG	Shell No.: 5	Inside wall temperature
16	B1H1.B1H1_S5_MID_TEMP_60SEC	B1H1_S5 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 5	Middle wall temperature
17	B1H1.B1H1_S5_PRESS_60SEC	B1H1_S5 PRESSURE 60 SECOND LAG	Shell No.: 5	Steam pressure
18	B1H1.B1H1_S6_INNER_TEMP_60SEC	B1H1_S6 INNER METAL TEMP 60 SECOND LAG	Shell No.: 6	Inside wall temperature
19	B1H1.B1H1_S6_MID_TEMP_60SEC	B1H1_S6 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 6	Middle wall temperature
20	B1H1.B1H1_S6_PRESS_60SEC	B1H1_S6 PRESSURE 60 SECOND LAG	Shell No.: 6	Steam pressure
21	B1H1.B1H1_S7_INNER_TEMP_60SEC	B1H1_S7 INNER METAL TEMP 60 SECOND LAG	Shell No.: 7	Inside wall temperature
22	B1H1.B1H1_S7_MID_TEMP_60SEC	B1H1_S7 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 7	Middle wall temperature
23	B1H1.B1H1_S7_PRESS_60SEC	B1H1_S7 PRESSURE 60 SECOND LAG	Shell No.: 7	Steam pressure
24	B1H1.B1H1_S8_INNER_TEMP_60SEC	B1H1_S8 INNER METAL TEMP 60 SECOND LAG	Shell No.: 8	Inside wall temperature
25	B1H1.B1H1_S8_MID_TEMP_60SEC	B1H1_S8 MIDDLE METAL TEMP 60 SECOND LAG	Shell No.: 8	Middle wall temperature
26	B1H1.B1H1_S8_PRESS_60SEC	B1H1_S8 PRESSURE 60 SECOND LAG	Shell No.: 8	Steam pressure
27	B1H1.B1H1_S9_INNER_TEMP_60SEC	B1H1_S9 INNER METAL TEMP 60 SECOND LAG	Shell No.: 9	Inside wall temperature

Live2csv Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.4.2 DXT Mapping

The DXT Mapping screen displays Tag Name Mapping and parameter association for variables coming from the DCS to the BSM software.



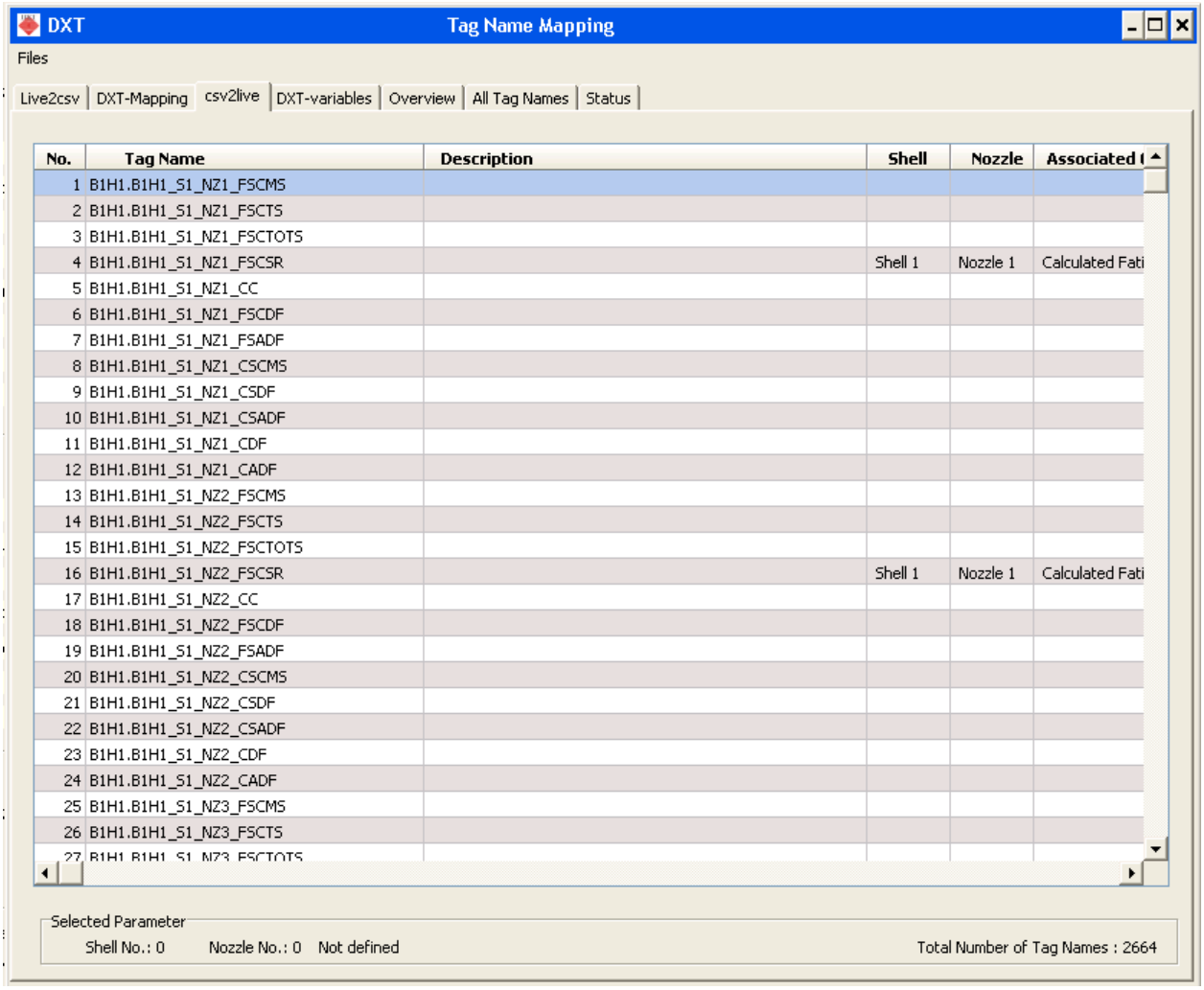
Name	Shell	Parameter	Value	Date	Time
B1H1.B1H1_S1_AW_INNER_TEMP_60SEC	Shell No.1	Inner wall temperature above water line	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S1_AW_MID_TEMP_60SEC	Shell No.1	Middle wall temperature above water line	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S1_BW_MID_TEMP_60SEC	Shell No.1	Middle wall temperature below water line	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S1_BW_INNER_TEMP_60SEC	Shell No.1	Inner wall temperature below water line	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S1_PRESS_60SEC	Shell No.1	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S2_INNER_TEMP_60SEC	Shell No.2	Inside wall temperature	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S2_MID_TEMP_60SEC	Shell No.2	Middle wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S2_PRESS_60SEC	Shell No.2	Steam pressure	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S3_INNER_TEMP_60SEC	Shell No.3	Inside wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S3_MID_TEMP_60SEC	Shell No.3	Middle wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S3_PRESS_60SEC	Shell No.3	Steam pressure	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S4_INNER_TEMP_60SEC	Shell No.4	Inside wall temperature	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S4_MID_TEMP_60SEC	Shell No.4	Middle wall temperature	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S4_PRESS_60SEC	Shell No.4	Steam pressure	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S5_INNER_TEMP_60SEC	Shell No.5	Inside wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S5_MID_TEMP_60SEC	Shell No.5	Middle wall temperature	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S5_PRESS_60SEC	Shell No.5	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S6_INNER_TEMP_60SEC	Shell No.6	Inside wall temperature	0.000	2013-05-10	15:12:49.560
B1H1.B1H1_S6_MID_TEMP_60SEC	Shell No.6	Middle wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S6_PRESS_60SEC	Shell No.6	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S7_INNER_TEMP_60SEC	Shell No.7	Inside wall temperature	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S7_MID_TEMP_60SEC	Shell No.7	Middle wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S7_PRESS_60SEC	Shell No.7	Steam pressure	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S8_INNER_TEMP_60SEC	Shell No.8	Inside wall temperature	0.000	2013-05-10	15:12:49.800
B1H1.B1H1_S8_MID_TEMP_60SEC	Shell No.8	Middle wall temperature	0.000	2013-05-10	15:12:49.640
B1H1.B1H1_S8_PRESS_60SEC	Shell No.8	Steam pressure	0.000	2013-05-10	15:12:49.720
B1H1.B1H1_S9_INNER_TEMP_60SEC	Shell No.9	Inside wall temperature	0.000	2013-05-10	15:12:49.640

DXT Mapping Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.4.3 Csv2live

The Csv2live screen displays the status for the Tag Name mapping of parameters and values going from the BSM software to the DCS.

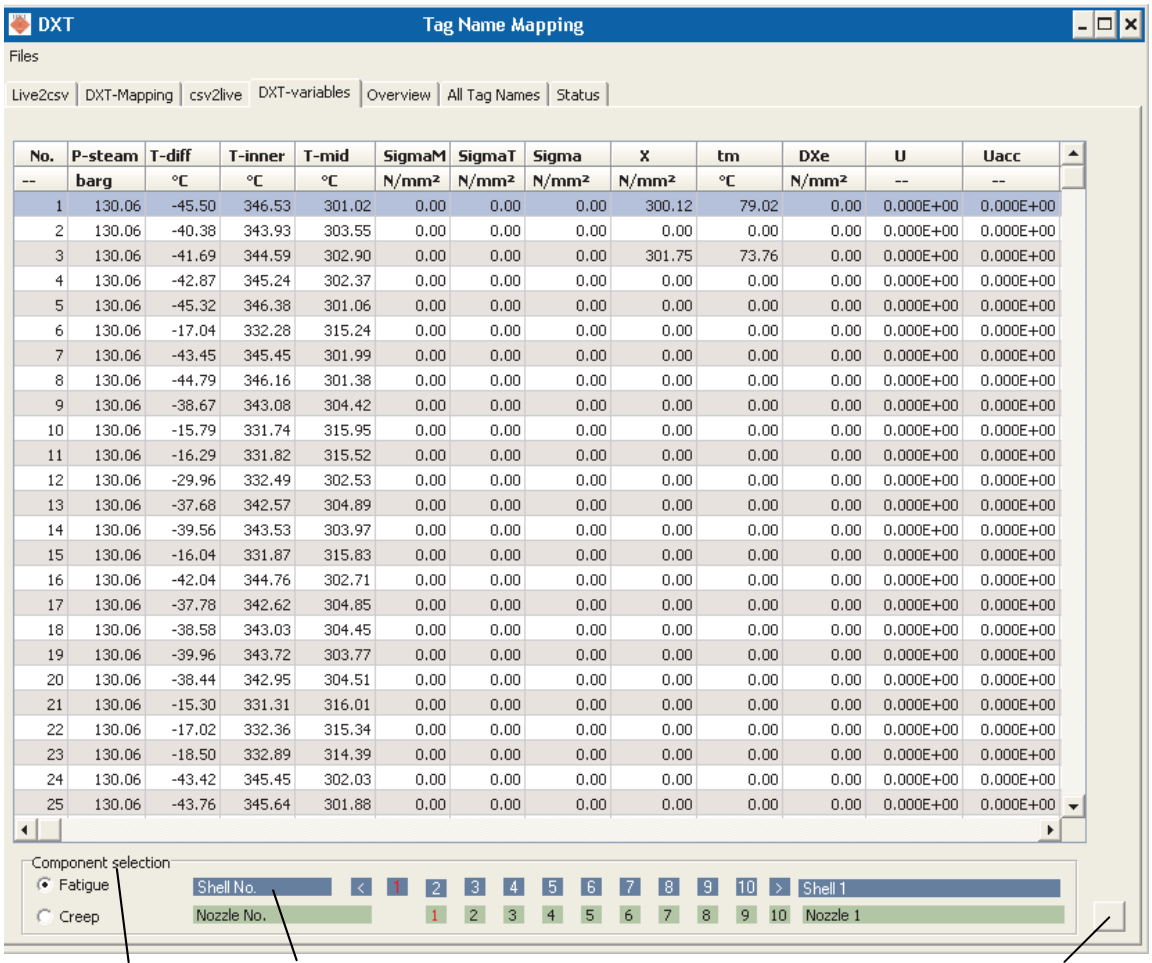


Csv2live Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.4.4 DXT–Variables

The DXT-Variables screen displays parameters used for fatigue and creep damage calculation for all monitored nozzle shells. The grid contains values for the latest 100 calculations and is updated every time the data exchange between the DCS and BSM software takes place. Individual nozzles on the monitored components can be selected in the Component Selection menu. The **Fatigue** and **Creep** buttons display or hide fatigue and creep parameters. Placing the cursor at the column headers provides a more detailed explanation and engineering units of the parameters listed in the grid.



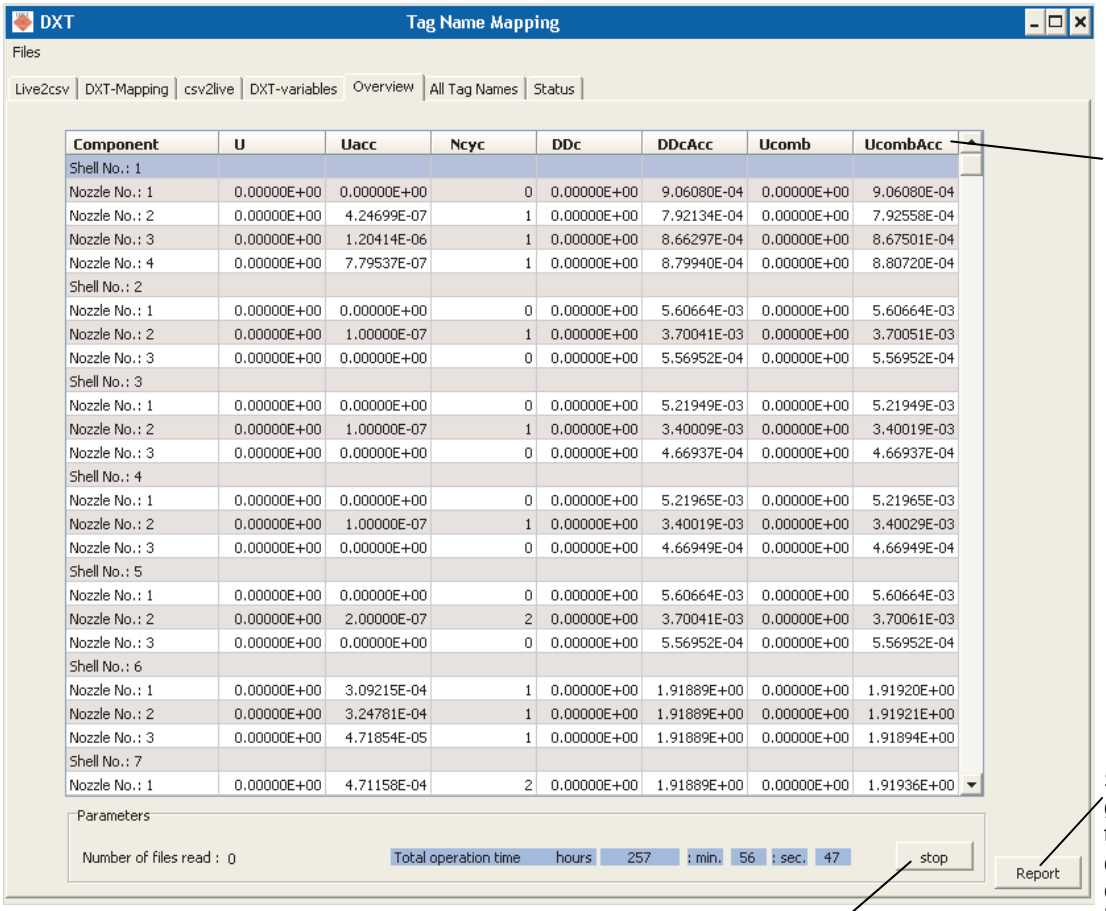
No.	P-steam	T-diff	T-inner	T-mid	SigmaM	SigmaT	Sigma	X	tm	DXe	U	Uacc
--	barg	°C	°C	°C	N/mm²	N/mm²	N/mm²	N/mm²	°C	N/mm²	--	--
1	130.06	-45.50	346.53	301.02	0.00	0.00	0.00	300.12	79.02	0.00	0.000E+00	0.000E+00
2	130.06	-40.38	343.93	303.55	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
3	130.06	-41.69	344.59	302.90	0.00	0.00	0.00	301.75	73.76	0.00	0.000E+00	0.000E+00
4	130.06	-42.87	345.24	302.37	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
5	130.06	-45.32	346.38	301.06	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
6	130.06	-17.04	332.28	315.24	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
7	130.06	-43.45	345.45	301.99	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
8	130.06	-44.79	346.16	301.38	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
9	130.06	-38.67	343.08	304.42	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
10	130.06	-15.79	331.74	315.95	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
11	130.06	-16.29	331.82	315.52	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
12	130.06	-29.96	332.49	302.53	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
13	130.06	-37.68	342.57	304.89	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
14	130.06	-39.56	343.53	303.97	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
15	130.06	-16.04	331.87	315.83	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
16	130.06	-42.04	344.76	302.71	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
17	130.06	-37.78	342.62	304.85	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
18	130.06	-38.58	343.03	304.45	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
19	130.06	-39.96	343.72	303.77	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
20	130.06	-38.44	342.95	304.51	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
21	130.06	-15.30	331.31	316.01	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
22	130.06	-17.02	332.36	315.34	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
23	130.06	-18.50	332.89	314.39	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
24	130.06	-43.42	345.45	302.03	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00
25	130.06	-43.76	345.64	301.88	0.00	0.00	0.00	0.00	0.00	0.00	0.000E+00	0.000E+00

DXT-Variables Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.4.5 Overview Tab

The Overview tab displays the calculated values for instant and accumulated fatigue and creep damage factions. Reports can be generated containing the fatigue and creep damage factions being monitored. The grid displays the latest 100 calculations. Fatigue damage is based on the last 100 cycles and the creep damage is accumulated from the last time the software was started to the present. Accumulated values are stored in the plant data historian.



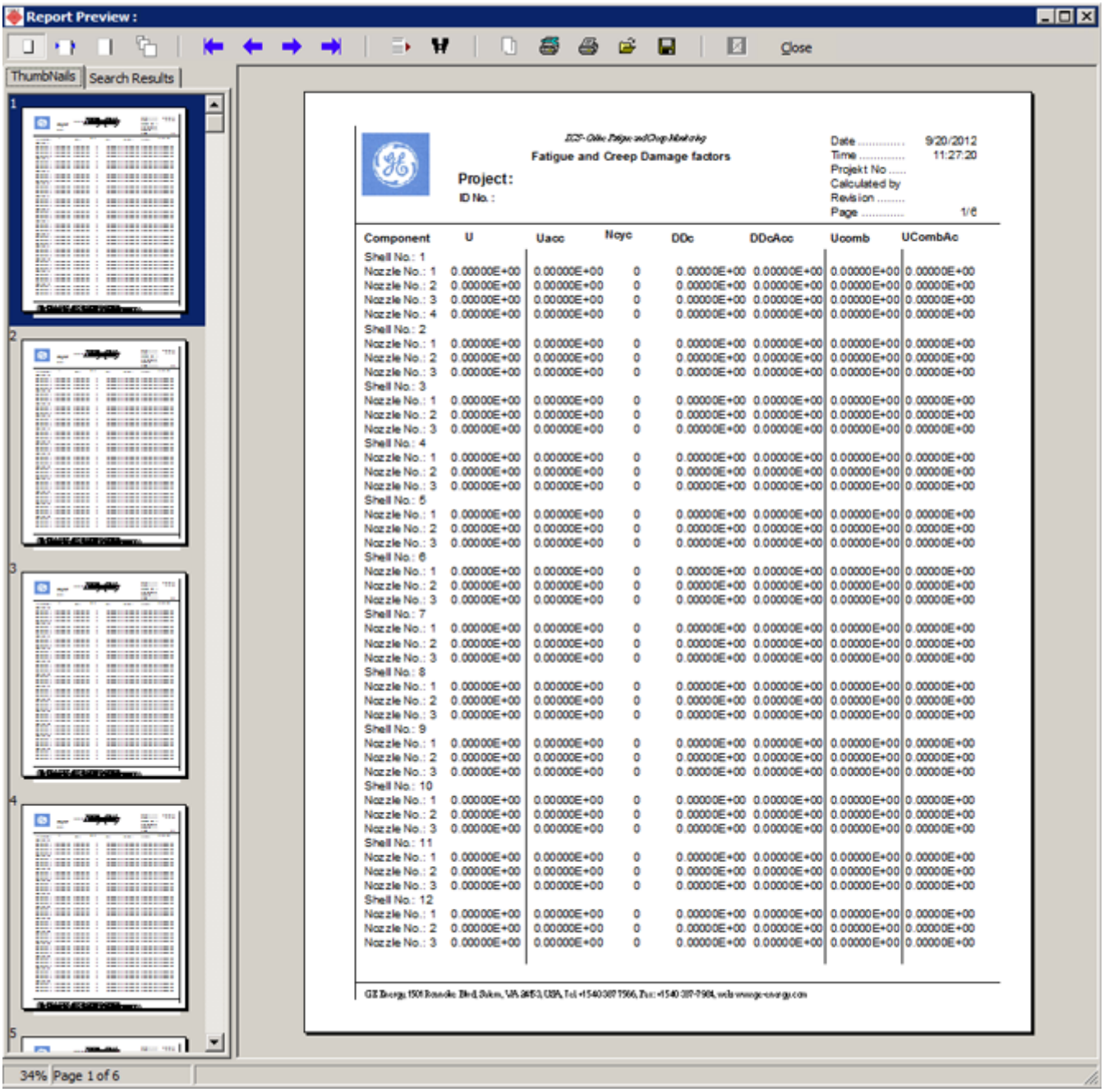
Placing the cursor at the column header provides a more detailed explanation and engineering units of the parameters listed in the grid.

Select **Report** to generate a report of the fatigue and creep damage fraction for the component nozzles being monitored.

Select **Stop** to suspend the parameter update.

Overview Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

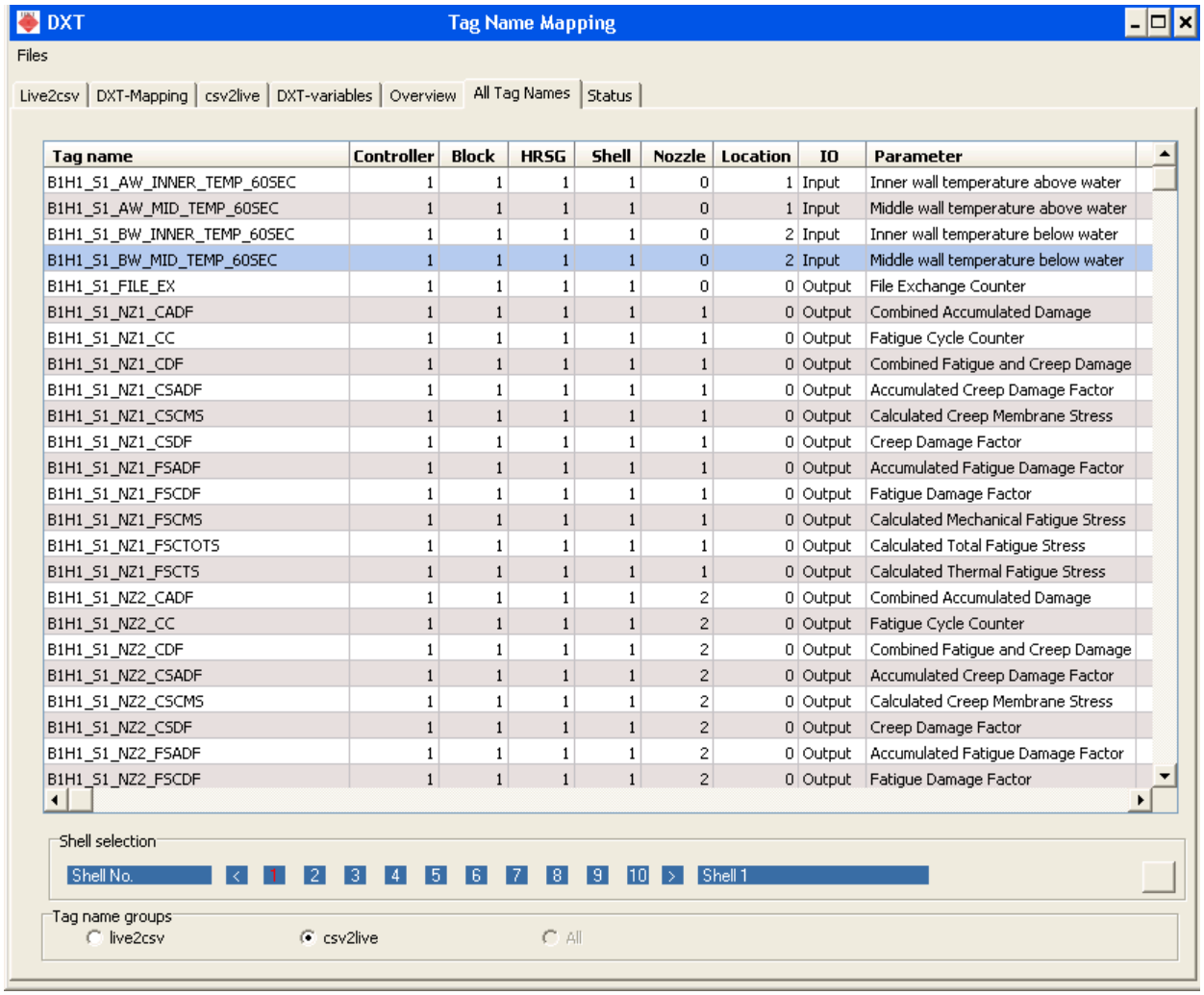


Example of Report Preview

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.4.6 All Tag Names Tab

The All Tag Names tab displays a list of all parameters and variables with attributes consumed and produced by the software, site block, HRSG, shells, and nozzles. The attributes identify the component and sub-component, if applicable, with which the parameter or variable is associated, whether it is an input (live2csv) or output (csv2live), and a description of the parameter or variable. Inputs and outputs from the BSM can be selected from the Tag name group menu.



Tag name	Controller	Block	HRSG	Shell	Nozzle	Location	IO	Parameter
B1H1_S1_AW_INNER_TEMP_60SEC	1	1	1	1	0	1	Input	Inner wall temperature above water
B1H1_S1_AW_MID_TEMP_60SEC	1	1	1	1	0	1	Input	Middle wall temperature above water
B1H1_S1_BW_INNER_TEMP_60SEC	1	1	1	1	0	2	Input	Inner wall temperature below water
B1H1_S1_BW_MID_TEMP_60SEC	1	1	1	1	0	2	Input	Middle wall temperature below water
B1H1_S1_FILE_EX	1	1	1	1	0	0	Output	File Exchange Counter
B1H1_S1_NZ1_CADF	1	1	1	1	1	0	Output	Combined Accumulated Damage
B1H1_S1_NZ1_CC	1	1	1	1	1	0	Output	Fatigue Cycle Counter
B1H1_S1_NZ1_CDF	1	1	1	1	1	0	Output	Combined Fatigue and Creep Damage
B1H1_S1_NZ1_CSADF	1	1	1	1	1	0	Output	Accumulated Creep Damage Factor
B1H1_S1_NZ1_CSCMS	1	1	1	1	1	0	Output	Calculated Creep Membrane Stress
B1H1_S1_NZ1_CSDF	1	1	1	1	1	0	Output	Creep Damage Factor
B1H1_S1_NZ1_FSADF	1	1	1	1	1	0	Output	Accumulated Fatigue Damage Factor
B1H1_S1_NZ1_FSCDF	1	1	1	1	1	0	Output	Fatigue Damage Factor
B1H1_S1_NZ1_FSCMS	1	1	1	1	1	0	Output	Calculated Mechanical Fatigue Stress
B1H1_S1_NZ1_FSCTOTS	1	1	1	1	1	0	Output	Calculated Total Fatigue Stress
B1H1_S1_NZ1_FSCTS	1	1	1	1	1	0	Output	Calculated Thermal Fatigue Stress
B1H1_S1_NZ2_CADF	1	1	1	1	2	0	Output	Combined Accumulated Damage
B1H1_S1_NZ2_CC	1	1	1	1	2	0	Output	Fatigue Cycle Counter
B1H1_S1_NZ2_CDF	1	1	1	1	2	0	Output	Combined Fatigue and Creep Damage
B1H1_S1_NZ2_CSADF	1	1	1	1	2	0	Output	Accumulated Creep Damage Factor
B1H1_S1_NZ2_CSCMS	1	1	1	1	2	0	Output	Calculated Creep Membrane Stress
B1H1_S1_NZ2_CSDF	1	1	1	1	2	0	Output	Creep Damage Factor
B1H1_S1_NZ2_FSADF	1	1	1	1	2	0	Output	Accumulated Fatigue Damage Factor
B1H1_S1_NZ2_FSCDF	1	1	1	1	2	0	Output	Fatigue Damage Factor

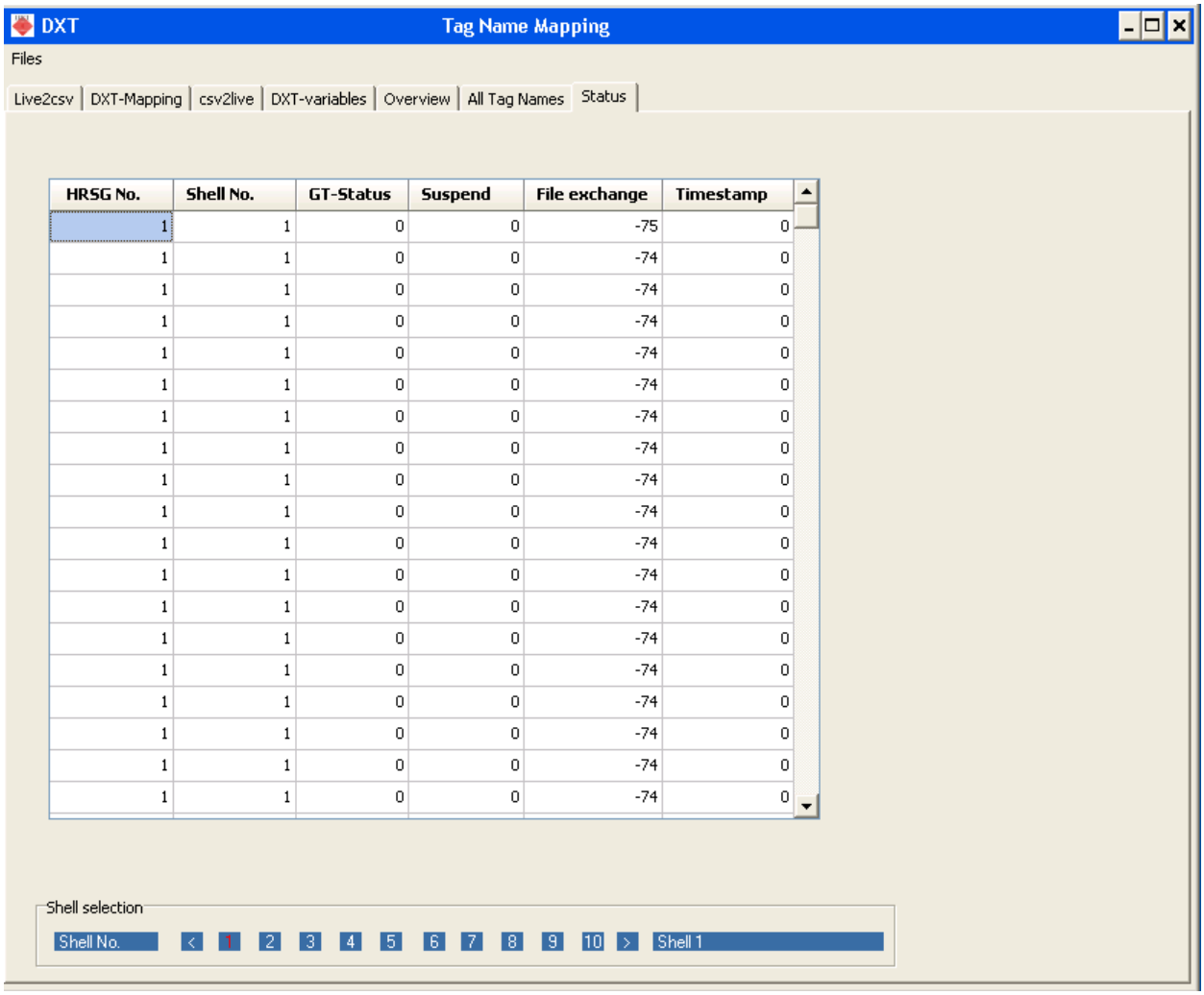
All Tag Names Screen

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.4.7 Status Tab

Refer to the section, Data Transfer Between the Controller and DXT calculation Engine, for a matrix of the suspend parameter.

The Status tab displays a list of variables and their status as related to communications with the Mark VIe DCS. The GT-Status represents the running state of the gas turbine for the particular HRSG identified and is consumed by the online software. The Suspend variable, also consumed by the online software, is a command from the Mark VIe DCS for the halt of calculations for that particular shell of that particular HRSG. The File-exchange variable is produced by the online software and represents feedback from the online software to the Mark VIe DCS on the status of the calculations for the particular shell of a particular HRSG.

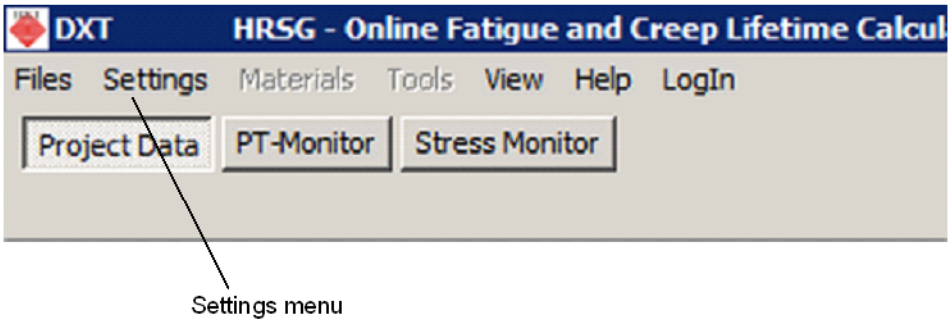


Status Screen

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

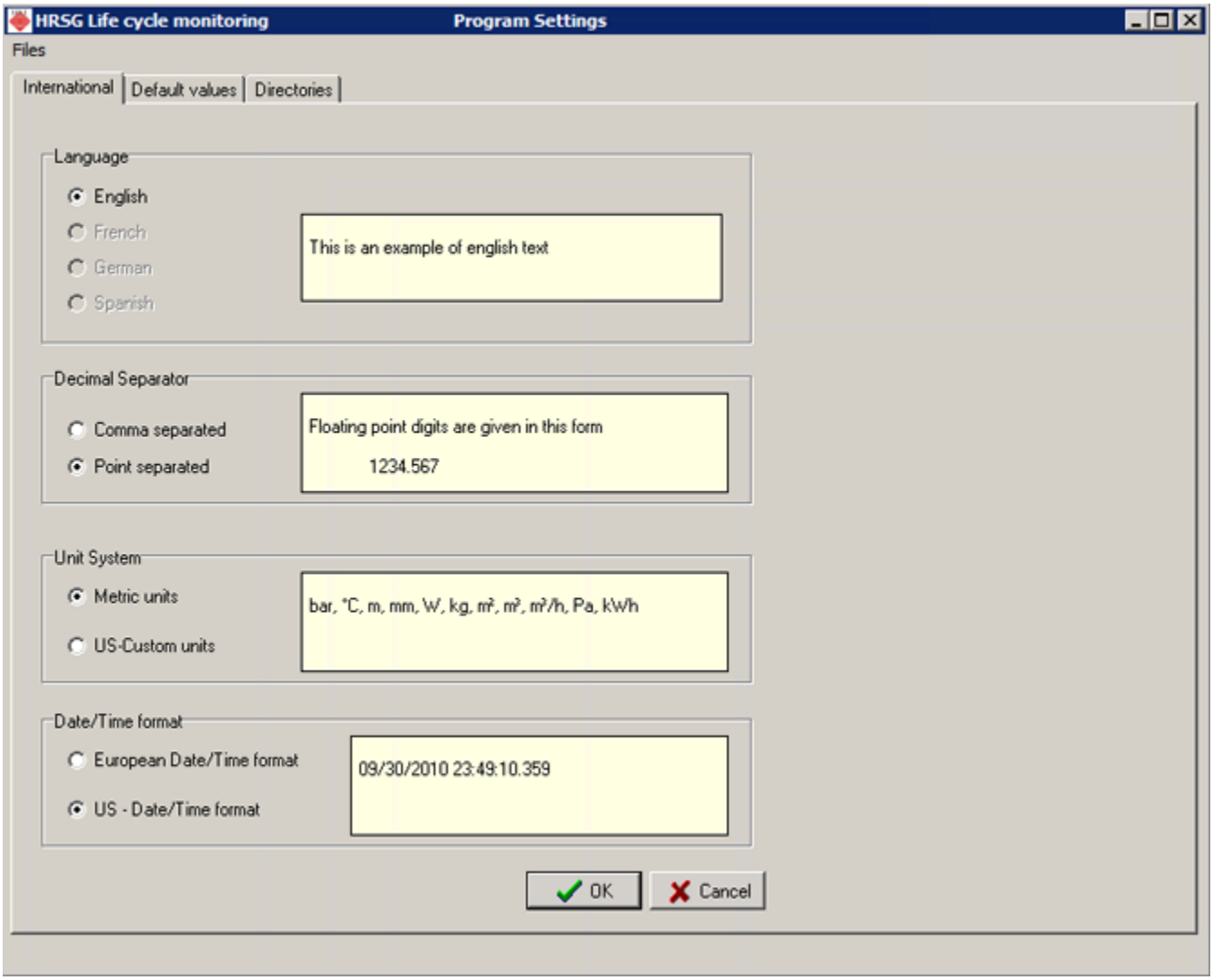
3.5 Program Settings

Program settings can be changed and stored from the Settings menu. Users can select either the International, Default values, or Directories screens.



3.5.1 International Tab

The International tab provides options for controls language display and data formatting, such as decimal separator, unit display, and time/date format.



International Screen

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

The following sections are displayed on the International screen:

- **Language** is available in English only. This setting cannot be modified in Monitor mode.
- **Decimal Separator** determines formatting used when reading or writing floating point values to the screen. It is important that the decimal separator is set equal to the decimal separator in the Windows operating system (OS) to correctly read values from the screen. This setting cannot be modified in Monitor mode.
- **Unit Split** toggles between Metric and US-Custom engineering unit for all parameters in the user interface. The measurement unit can be changed by selecting the units and pressing **OK**.

This does not apply to Sabiya.

- **Date/Time Format** sets the date and time format for the time stamp exchange between the DCS and BSM software when using .csv files for data exchange. It is important that the date/time format is set equal to the date/time format used in Windows OS. The date/time setting cannot be modified in the Monitor mode. When using the OPC server for data exchange, the time stamp comes directly from the DCS in ms since January 1, 1900. In this case, the actual date/time format does not matter.

  The following abbreviations are used:

  - **dd:** day in month
  - **mm:** month
  - **yyyy:** year
  - **hh:** hour (in 24–hour format)
  - **nn:** minutes
  - **ss:** seconds

Language Display Date/Time Format

Language	Date/Time Format	Example
European	dd-mm-yyyy hh:nn:ss	08-10-2012 14:22:50
US	mm/dd/yyyy hh:nn:ss	10/08/2012 14:22:50

➢ To apply settings

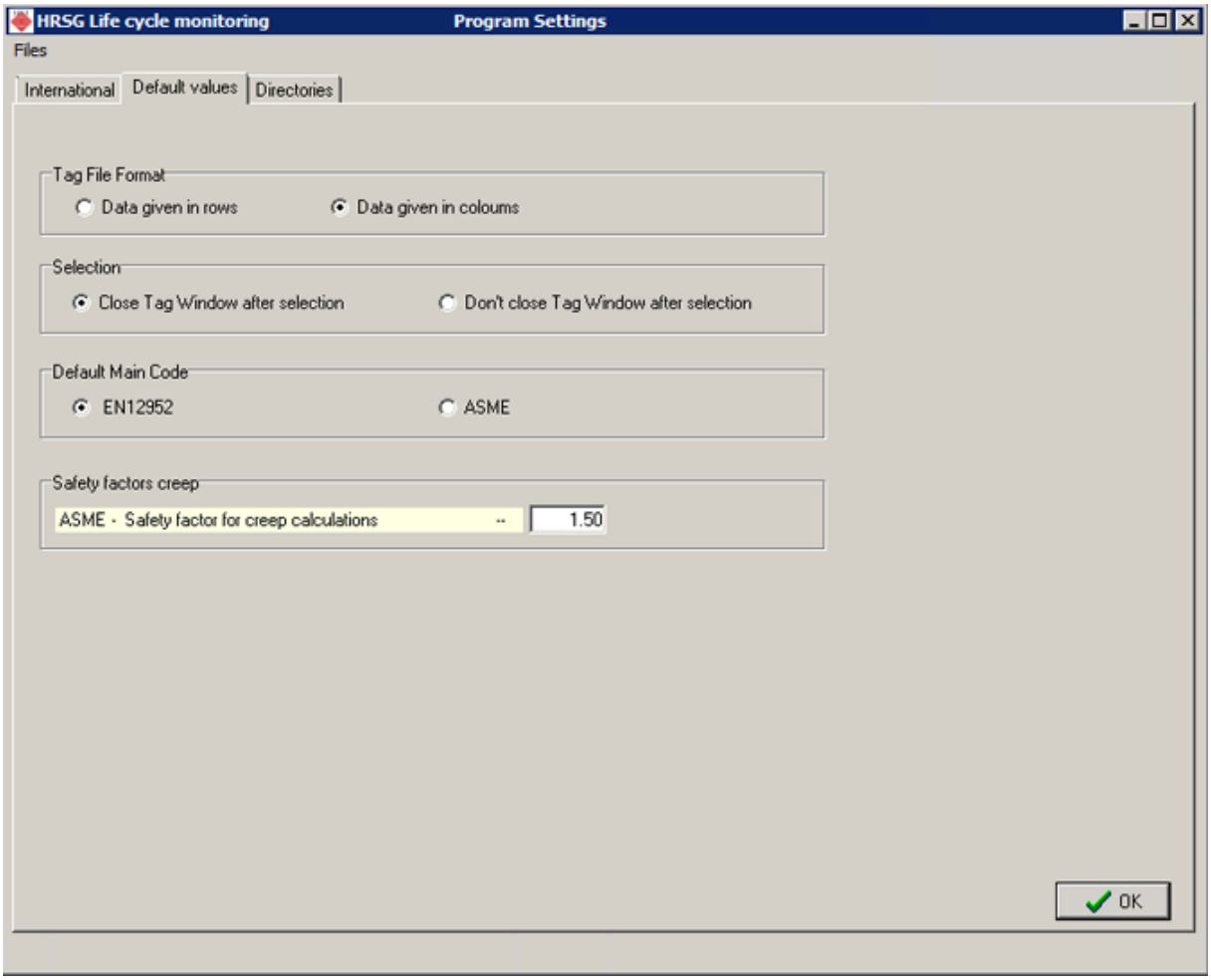
1. Click **OK** to immediately activate the settings.
2. From the **File** menu, select **Save Setting** to save the settings. The settings are saved as default values for the next time the software runs.

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.5.2 Default Values Tab

The Default Values tab can only be accessed in Full or Partial Access modes.

The Default Values tab enables settings for data communication between the DCS, BSM software, and the calculation model for fatigue and creep damage.



Default Values screen

The following sections are displayed on the Default Values screen:

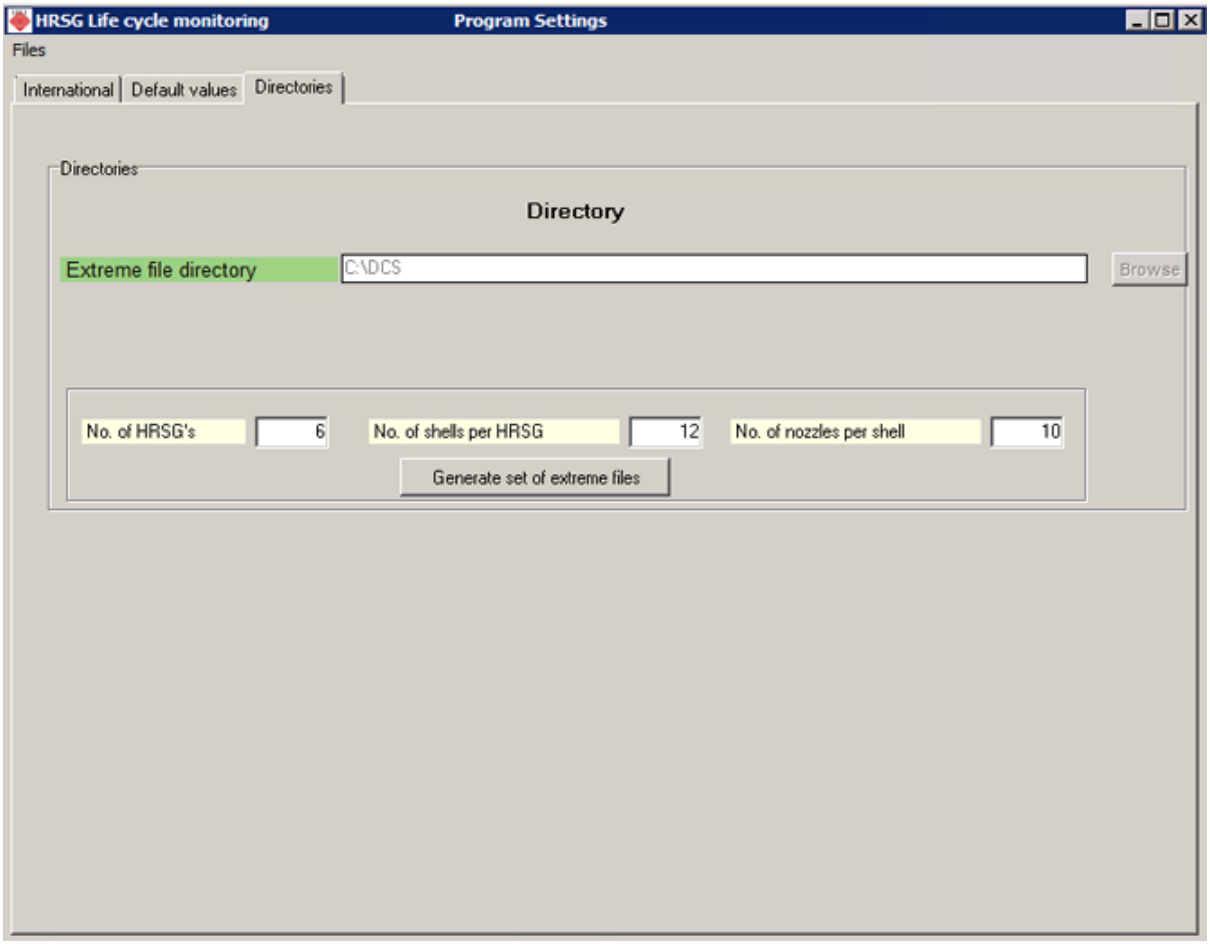
- **Tag File Format** specifies the file format for the .csv files used for data exchange between the DCS and BSM software.
- **Selection** determines the behavior of the user interface when the Show Tag Name page is selected. The Selection menu is not implemented in this release of the BSM software.
- **Default Main Code** determines the default calculation standard for fatigue and creep damage calculation. Currently, only EN12952 methods are implemented in this release of the BSM software.
- **ASME-Safety factor for creep calculation** specifies the safety factor to be used when deriving the material allowable stress values in the creep range when the time dependent creep rupture strength is known for use in the ASME analysis methodology. ASME methodology is not available in this release of the BSM software.

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

3.5.3 Directories Tab

The Directories tab can only be accessed in Full or Partial Access modes.

The Directories tab defines a set of *extreme* files. Extreme files are text files that contain process and stress parameters for identified extreme points for every nozzle used in the cycle counting method. A set of extreme files must be declared *before* running the BSM software. The number and naming of the files depend on the given numbers of HRSGs, shells, and nozzles. The extreme files are used for troubleshooting during project setup and should not be modified by the user.



Directories Screen

- **To generate a set of extreme files:** from the directory specified under **Extreme file directory**, click **Generate set of extreme files**.

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

The following values are written out for every detected extreme point (refer to EN12952-4):

- Steam pressure
- Steam temperature
- Inner wall temperature
- Middle wall temperature
- Mechanical stress value
- Thermal stress value
- Total stress value
- Extreme stress values for cycle detection
- Stress range for detected cycle
- Middle wall temperatures for operation point used in cycle detection.
- Fatigue damage factor for detected cycle
- Number of cycles detected with a stress range below the threshold for fatigue damage to occur

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Notes

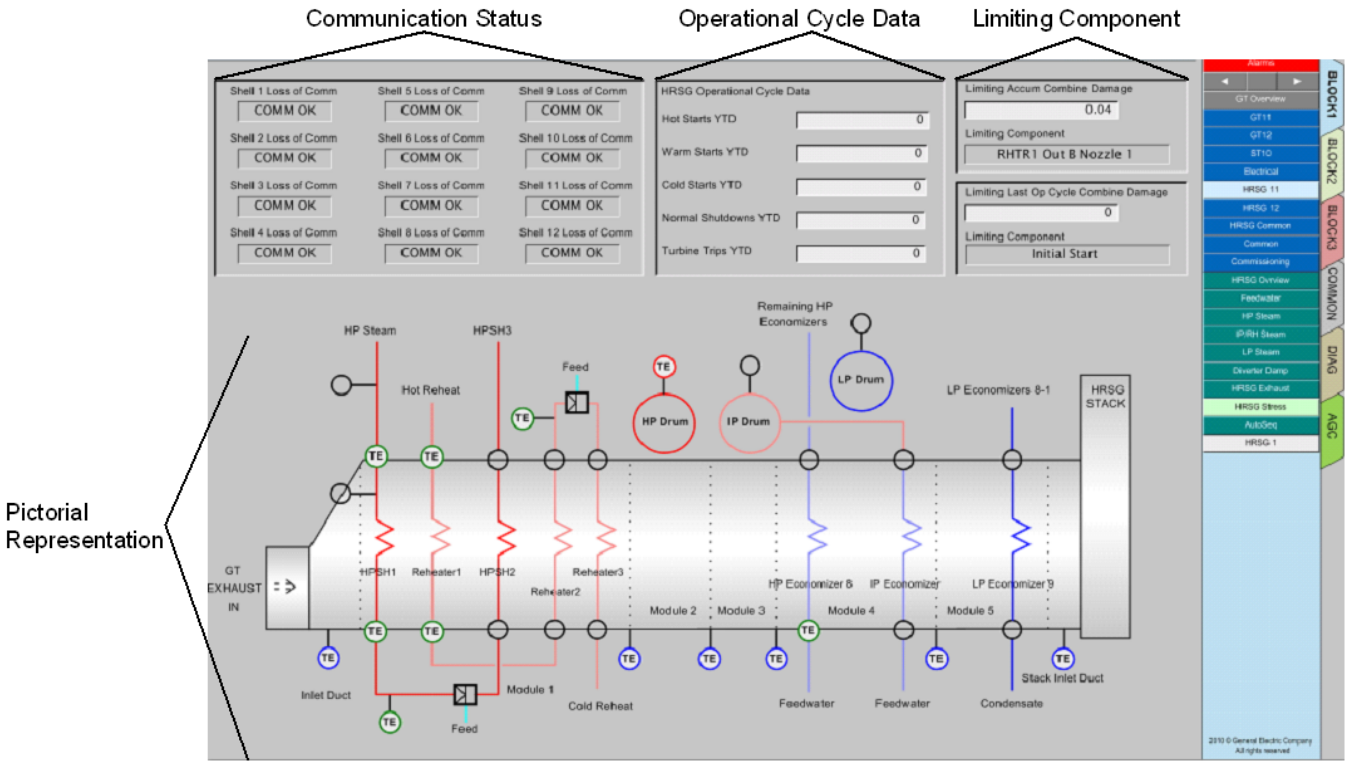
GE Proprietary and Internal (Class II) – This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.

4 HMI Screens and Operator Interface

Each HRSG has a dedicated BSM screen.

The BSM product requires no operator interaction to function. However, screens are provided for system monitoring. The operator interface screen consists of four zones:

- *Communication status*
- *Cumulative operational cycle data*
- *Limiting component* (contains identification and current values)
- *Pictorial representation with faceplates*



Block 1 - HRSG 11 HRSG Stress Screen, Showing Navigation

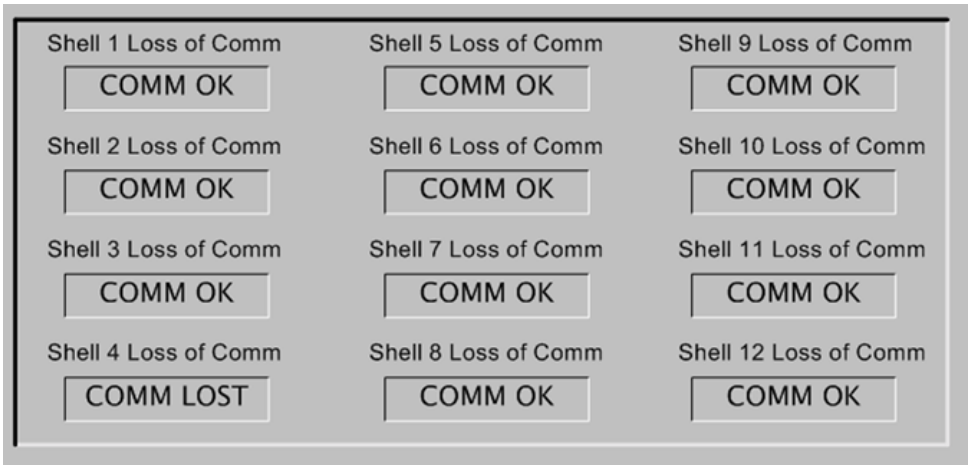
GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

4.1 Communication Status

Communication Status indicates the communication health between the Mark VIe control system and the online calculation engine. Communication is monitored at the controller level. This ensures that communications are being monitored from the online calculation engine to the controller level. Each component communication is monitored individually to ensure that all CSV files are functioning properly and the calculation engine has not suspended calculation on a particular component due to missing or out of range process signals.

Since the HRSGSTRESS_SVR's Scheduled Tasks control the communications and the cycle is set for one minute, the controller logic monitors for four consecutive lost updates by resetting a four minute timer every time new data is available. If no data is received in any four minute block, the timer will time out and declare a communications loss for that component. The loss of communication to all components could indicate:

- DXT online software has either crashed or locked up and requires restarting.
- HRSGSTRESS_SVR WorkstationST services have failed and require restarting.
- HRSGSTRESS_SVR computer has failed.
- Possible UDH network problem.

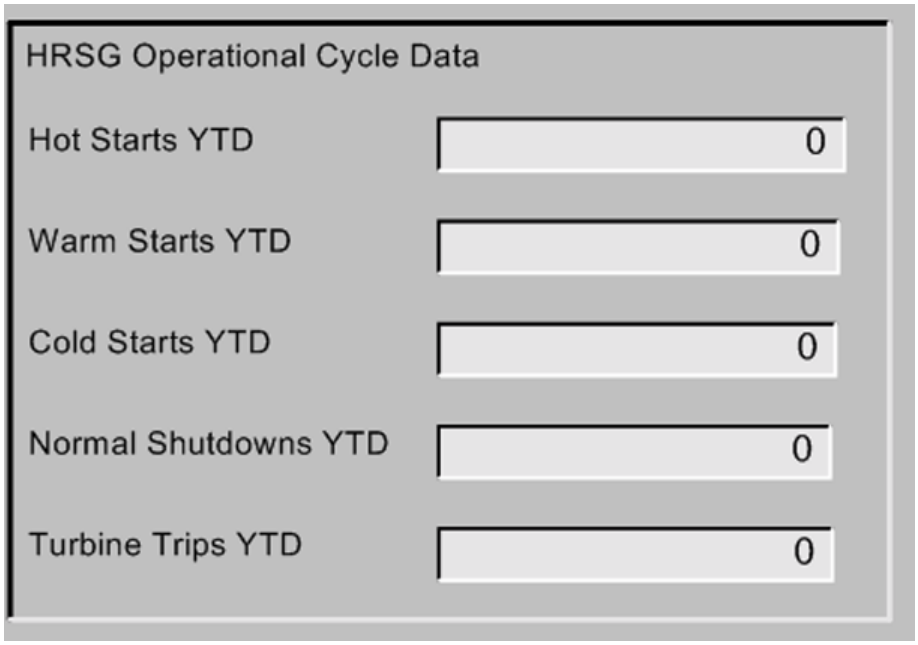


Communication Zone Indicating Shell 4 Is Not Communicating

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

4.2 HRSG Operational Cycle Data

Operational Cycle Data provides a quick report of the starts and shutdowns of the HRSG gas turbine. HRSG counters are calendar-based and reset at the start of a new year. The counters give an intuitive feel for the stresses the HRSG is subjected to when compared with the limiting damage.



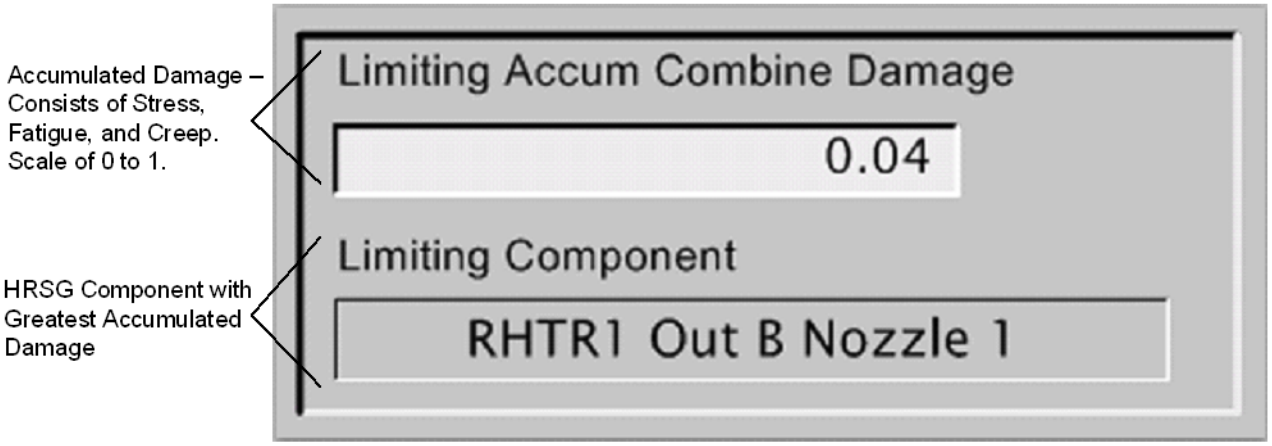
HRSG Operational Cycle Data

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

4.3 Limiting Component

The Limiting Component consists of two primary pieces of data:

- **Limiting Accumulated Combine Damage** displays the specific HRSG component with the maximum amount of combined damage with its running accumulation of stress, fatigue and creep damage to date.
- **Limiting Last Operational Cycle Combine Damage** displays the component that saw the greatest accumulated damage during the last gas turbine cycle from start up to shut down and the accumulated combined damage during the cycle.

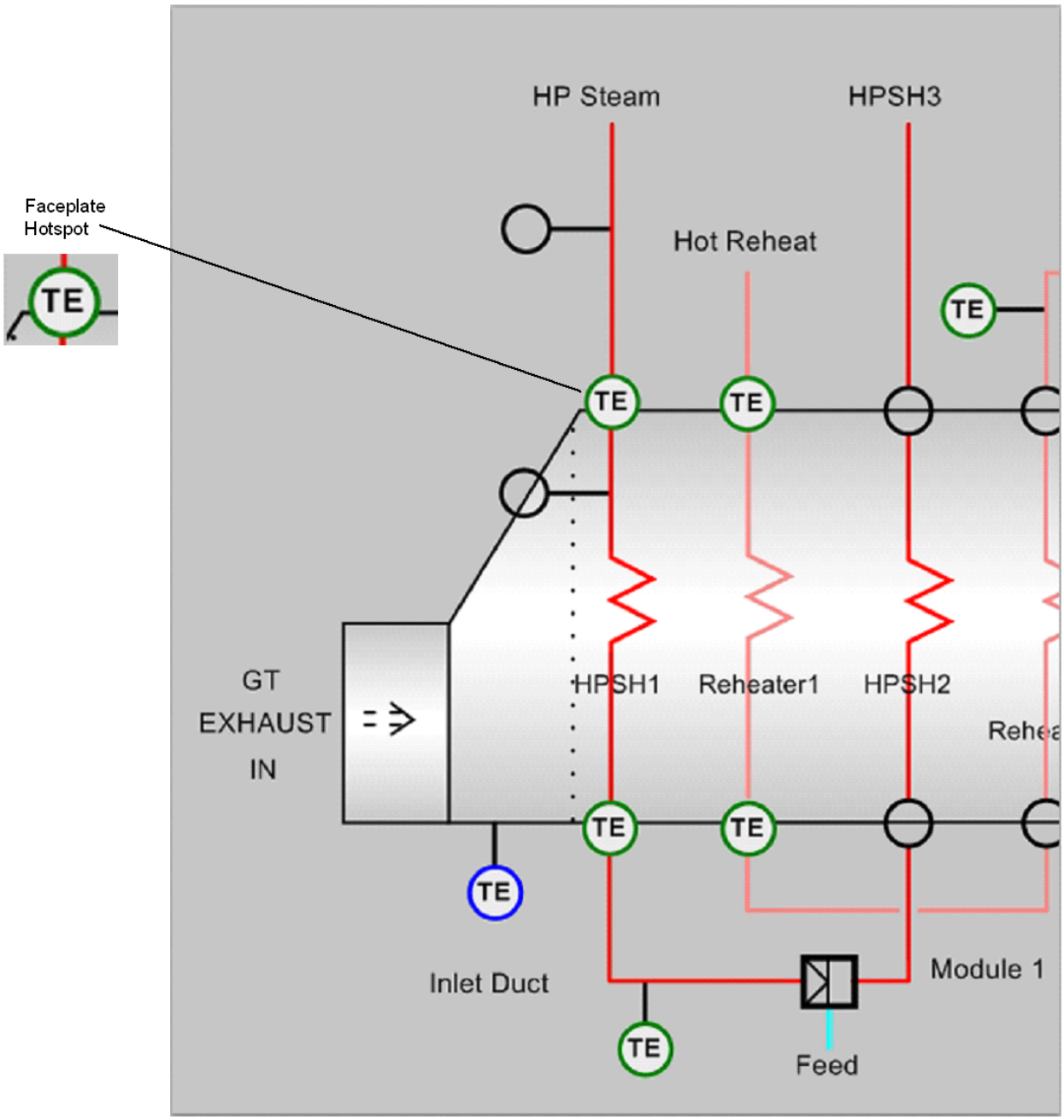


Limiting Component and the Accumulated Damage

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

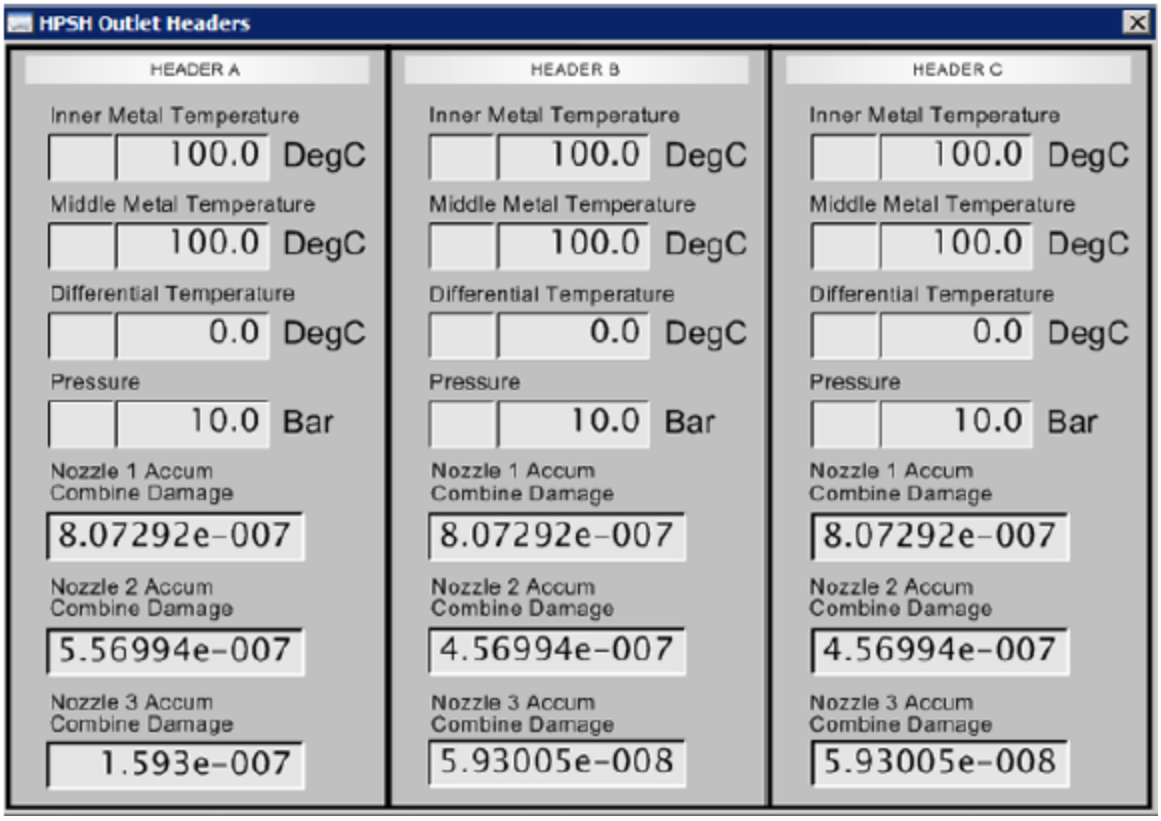
4.4 Faceplates

Each HRSG Stress graphic screen contains *hotspots* on the HRSG pictorial representation. Hotspots open faceplates that display critical data for each component, such as the process data containing temperatures, pressure, and accumulation for each monitored component nozzle.



HRSG Stress Graphic Screen With Hotspots

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



Standard Faceplate Displaying HP Super-Heater

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

5 Historical Data

Critical data for each monitored component is archived in the plant data historian, which includes the inner and mid-wall temperatures, drum or shell pressure, and the accumulated fatigue and creep for each nozzle of the shell. This data supports the user in developing trends for the accumulated shell damages, temperatures, and pressure to be overlaid with other plant process data, such as gas turbine exhaust gas temperature, steam flows, gas turbine watts, and so forth.

GE can provide consulting services to assist in the development of custom reports to meet site requirements.

Standardized reports can be created from the historian for operators' use in maintaining and evaluating plant condition. Since operational philosophy and metrics are not standard, GE does not provide any standard reports with the BSM system.

The BSM provides the following set naming convention for data archiving:

Convention 1 for process data:
{Controller Name}.B{block number}H{HRSG number}_S{shell number}_{descriptor}

For example:
B1H1.B1H1_S1_PRESS_60SEC is controller B1H1 - Block 1 - HRSG 1 - Shell 1 – PRESS_60SEC

Each tag ending in 60SEC indicates that the value of the tag has been processed through median or dual selects if the signal originates from either dual or triple transmitters, and the value has been passed through a 60 second averaging function.

Convention 2 for stress and creep values:
{Controller Name}.B{block number}H{HRSG number}_S{shell number}_N{nozzle number}_{descriptor}

For example:
B2H1.B2H1_S6_N2_CSADF_GE is controller B2H1 - Block 2 - HRSG 1 - Shell 6 – Nozzle 2 – Creep Stress Accumulated Damage Factor

Name	Description	Historian Deadband	Units
B1H1_S1_NZ1_CSADF_GE	B1H1_S1 Nozzle 1 Accumulated Creep Damage Factor GE	0	DF
B1H1_S1_NZ1_CADF_GE	B1H1_S1 Nozzle 1 Combine Accumulated Damage GE	0	DF
B1H1_S1_NZ1_FSADF_GE	B1H1_S1 Nozzle 1 Accumulated Fatigue Damage Factor GE	0	DF
B1H1_S1_NZ2_CSADF_GE	B1H1_S1 Nozzle 2 Accumulated Creep Damage Factorr GE	0	DF
B1H1_S1_NZ2_CADF_GE	B1H1_S1 Nozzle 2 Combine Accumulated Damage GE	0	DF
B1H1_S1_NZ2_FSADF_GE	B1H1_S1 Nozzle 2 Accumulated Fatigue Damage Factor GE	0	DF
B1H1_S1_NZ3_CSADF_GE	B1H1_S1 Nozzle 3 Accumulated Creep Damage Factor GE	0	DF
B1H1_S1_NZ3_CADF_GE	B1H1_S1 Nozzle 3 Combine Accumulated Damage GE	0	DF
B1H1_S1_NZ3_FSADF_GE	B1H1_S1 Nozzle 3Accumulated Fatigue Damage Factor GE	0	DF
B1H1_S1_AW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP ABOVE WATERLINE 60 SECOND LAG	0.1	DegC
B1H1_S1_AW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP ABOVE WATERLINE 60 SECOND LAG	0.1	DegC
B1H1_S1_BW_MID_TEMP_60SEC	B1H1_S1 MIDDLE WALL TEMP BELOW WATER 60 SECOND LAG	0.1	DegC
B1H1_S1_BW_INNER_TEMP_60SEC	B1H1_S1 INNER WALL TEMP BELOW WATER 60 SECOND LAG	0.1	DegC
B1H1_S1_PRESS_60SEC	B1H1_S1 60 SECOND LAG PRESSURE INPUT	0.1	Bar

Typical Historian Points for a Shell (Component)

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

Notes

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

6 Diagnostics and Alarms

Communications between the DCS and BSM program is monitored from each controller that transmits and receives data from the BSM. Sabiya has six controllers (one per HRSG) that produce process data and consume calculated data. Each controller monitors communication for each HRSG component being monitored by the DXT online calculation engine.

Each HRSG controller monitors twelve components (one drum and eleven shells). The BSM online calculation engine monitors for a change in the time stamp provided for each component by the DCS, such as a time stamp in the live2csv.csv files, if used, or through a UTC-based time stamp if the OPC DA server is used. When a time stamp changes, the BSM online calculation engine calculates and writes the output data to the csv2live.csv files, if used, or to the OPC DA file exchange variable, *FILE_EX*. A *FILE_EX* variable exists for each component. The value of *FILE_EX* changes every time the file is written. The controller monitors this tag for a change and, if a change is not detected for four minutes, declares communication is lost for that component.

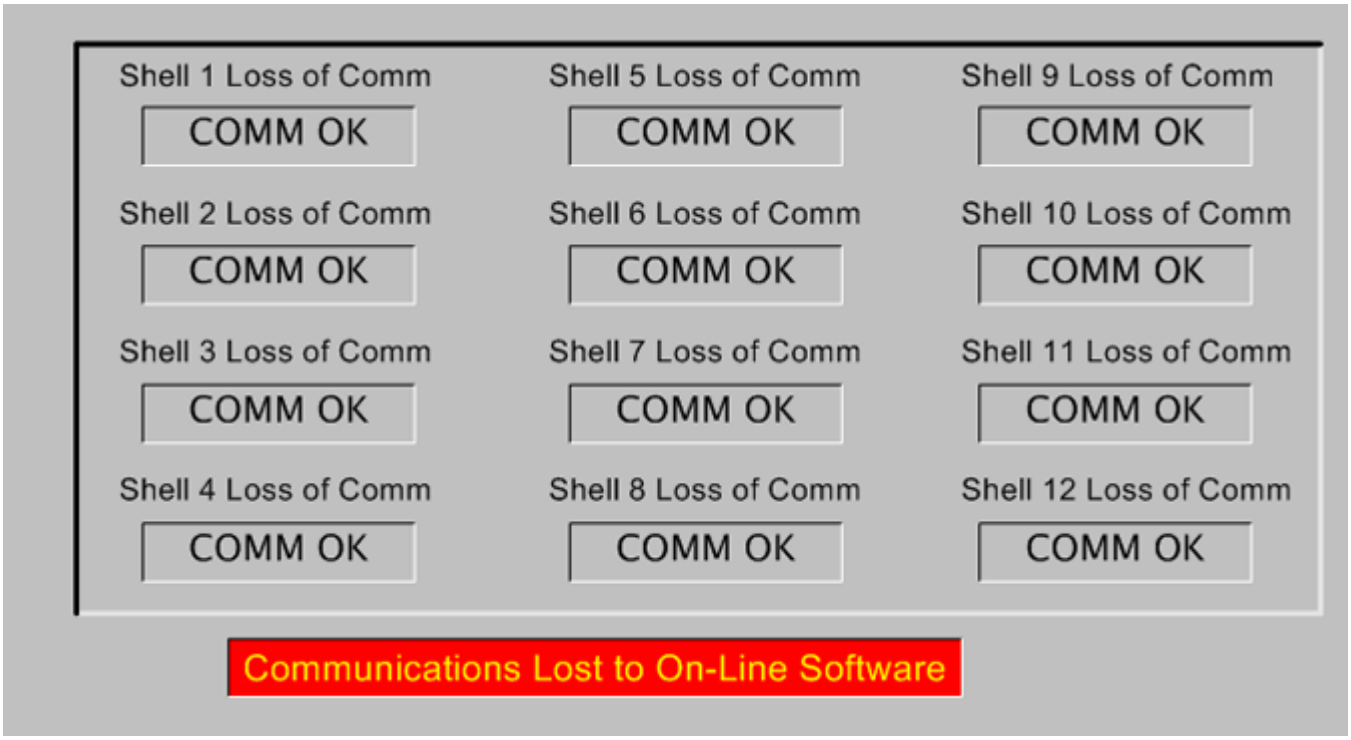
The controller generates a diagnostic alarm that is sent to the Alarm Server and changes the communication display status for that component on the HRSG Stress CIMPLICITY* screen. If the controller detects a loss of communications for every component being monitored, it declares a communication loss to the online software and generates a diagnostic alarm, which displays on the HRSG Stress CIMPLICITY screen and sent to the Alarm Server. If communication to all components in a controller is lost within 30 seconds of each other, the individual diagnostic alarm is blocked and only the loss of communication diagnostic alarm is sent to the Alarm Server. If the condition exists for four hours or longer, the active diagnostic alarms are re-generated.

Each component has a collection of counters internal to its DCS controller logic. These counters increment when any diagnostic or process alarms are generated and should be reviewed for assistance in troubleshooting when higher than expected damage factors occur and no operational causes are suspected.

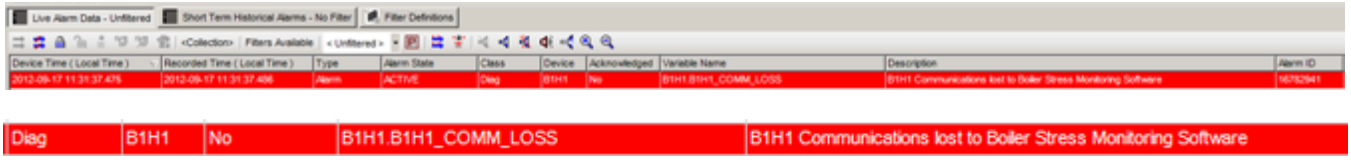
Typical faults that generate a Communications Loss alarm include:

- DXT online software failed
- HRSGSTRESS_SVR workstation hardware failed
- WorkstationST services failed on the HRSGSTRESS_SVR workstation
- Network communications lost between the workstation and the UDH network

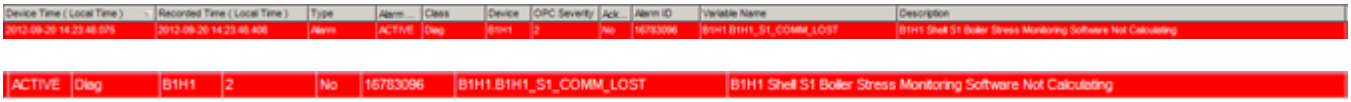
GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



Lost Communications Status on the HRSG Stress CIMPLICITY Screen



Diagnostic Communication Loss Alarming as displayed in the Alarm Server



Diagnostic Alarm for Loss of Communication for an Individual Component

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

6.1 Process Alarms

The HRSG controller monitors the thermocouple temperatures for the inner and middle metal temperatures for each component. Each component being monitored by the BSM system has been configured with a metallurgically-based high alarm, a Hi-Hi alarm, which is at the upper range of the design temperature for the component. A Hi-Hi alarm is generated when the component exceeds either the maximum allowable temperature at which the component should not be operated beyond, or the 20 mA value to which the temperature transmitter has been calibrated. A low temperature alarm is also provided. The normal setpoint is 5°C (41° F) when the HRSG is in operation and decreases to 0°C (32° F) when not in operation.

If a thermocouple input exceeds any alarm setpoint threshold, a Level 3 process alarm is generated by the HRSG controller, and the appropriate suspend command is passed to the BSM online software. The DXT online calculation engine suspends calculations involving the affected temperature. The Hi-Hi alarm will be re-generated every four hours until the temperature signal drops at least 5°C (41 °F) below the maximum allowable temperature. The status field for the affected temperature is updated for that component on the BSM screen. Bad Quality diagnostic alarms are passed to the Alarm Server and the CIMPLICITY screen.

Device Time (Local Time)	Recorded Time (Local Time)	Type	Alarm	Class	Device	OPC Severity	Ack	Alarm ID	Variable Name	Description
2012-09-20 14:26:54.475	2012-09-20 14:26:54.487	Alarm	H	LVL_3	B1H1	2	No	67115249	B1H1.11HAD70CT002AXQ01_A	B1H1 HP Drum Above Water Inner Metal Temperature has Exceeded the Maximum Allowable Hi

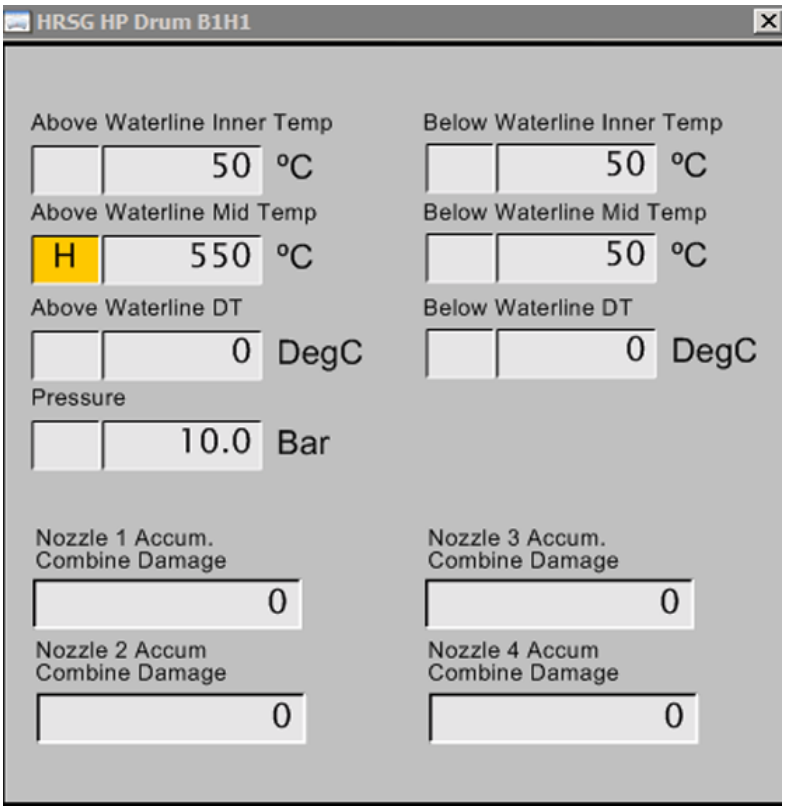
H	LVL_3	B1H1	2	No	67115249	B1H1.11HAD70CT002AXQ01_A	B1H1 HP Drum Above Water Inner Metal Temperature has Exceeded the Maximum Allowable Hi

High Temperature Alarm Sent to the Alarm Server

The following additional process alarms are provided through calculations:

- A Level 3 alarm is generated if the absolute value of the differential temperature between the inner and middle wall temperature of a component exceeds the expected maximum differential as calculated by the online software for that component. The differential temperature is calculated after signal filtering to ensure the alarm is not triggered by erroneous noise.

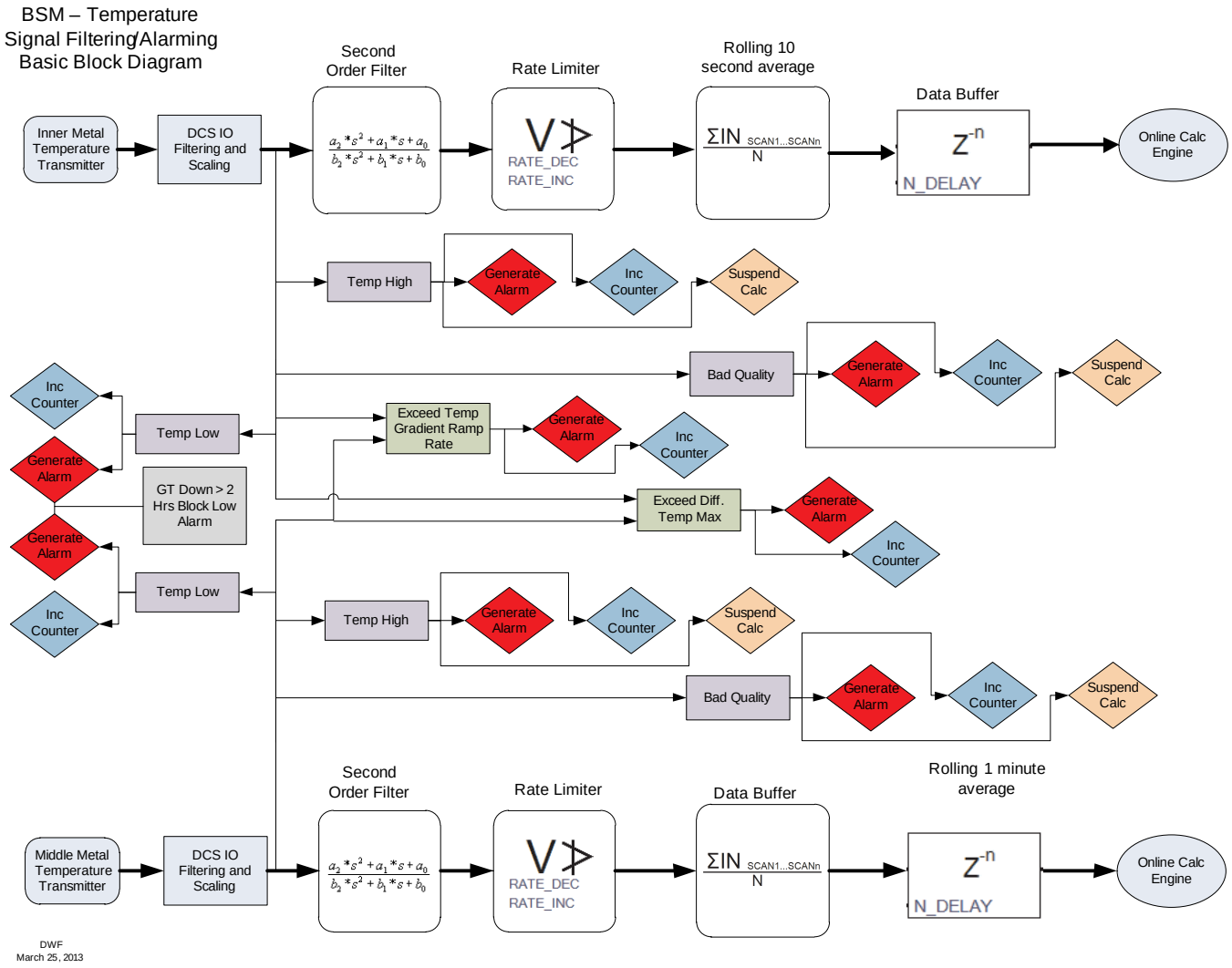
GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



Typical Temperature Alarm Status Indication on HRSG Stress CIMPLICITY Screen

- A Level 3 alarm is generated when the inner metal temperature of a component exceeds the maximum temperature gradient (temperature ramp in degC/minute) and may indicate that the component is being ramped faster than was originally designed to yield the designed lifetime of the component.

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



Block Diagram of Process Signal Processing

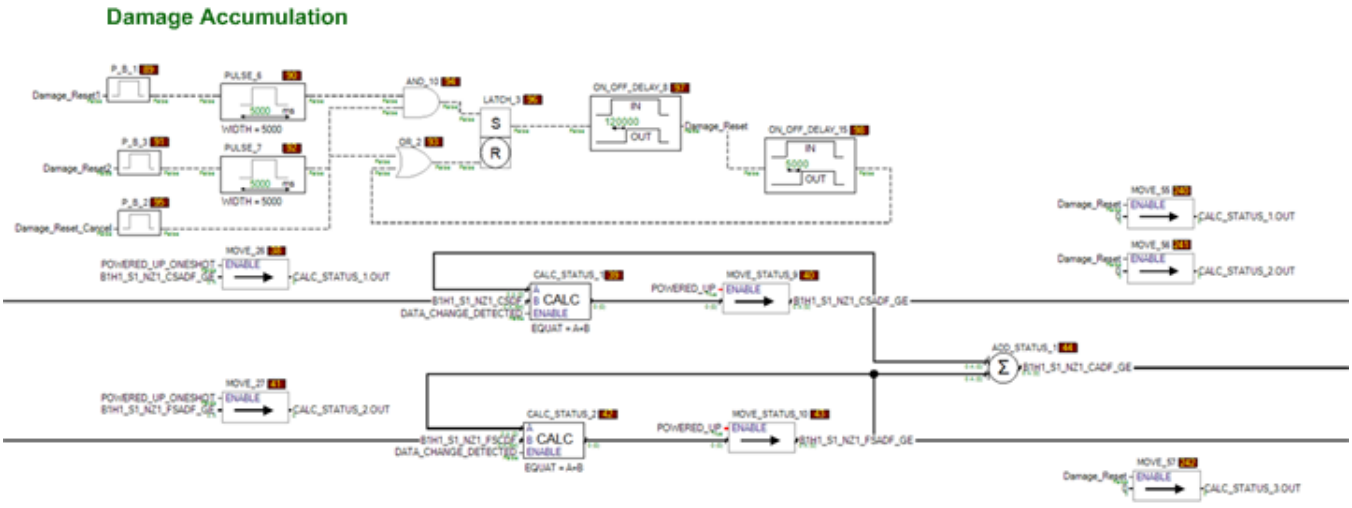
GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*

6.2 Component Damage Factors

The online calculation engine calculates a Creep damage factor and a Fatigue damage factor fraction for each critical nozzle of a component. The Creep damage is calculated from successive data updates which occurs every minute. The Creep damage number is added in the Mark VIe DCS to the previous total to maintain a cumulative Creep damage value. The Fatigue damage number is updated when the online calculation engine detects a completed fatigue cycle and updates the calculated Fatigue damage for the cycle to the Mark VIe DCS. These two values are stored in non-volatile ram (NOVRAM) of the HRSG controller to ensure protection of the numbers during a controller reboot. The two numbers are summed together to give the Combined Accumulated Damage Factor. This value is scaled 0 – 1 DF.

Proper access privileges are required to reset the Creep and Fatigue non-volatile numbers and modify data in the controller.

A reset mechanism is in place in the controller logic to reset the non-volatile Creep and Fatigue damage numbers as needed, such as in a replacement of a full component. The reset only affects the non-volatile number in the controller with the history of the values preserved in the historian. The toggling of two internal push-buttons within 5 seconds of each other sets a latch and starts a two minute timer. A reset cancel internal push-button is also provided to cancel the reset before the two minute timer times out, as needed. Once the numbers have been reset, the only mechanism to restore the previous numbers is to manually enter them into the Creep and Fatigue non-volatile online value based on the historical value.



Typical Reset Logic for Non-Volatile Creep and Fatigue Variables

GE Proprietary and Internal (Class II) – *This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*



1501 Roanoke Blvd.
Salem, VA 24153 USA

GE Proprietary and Internal (Class II) *– This document contains proprietary information of GE and is intended for internal use only. It may not be used, shown, reproduced, or disclosed outside of GE without the express written consent of GE.*